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HY Neuro, by Dr Michael D Mehlman

This PDF discusses with you HY diagnoses, risk factors, mechanisms, and even some neuro-related pharm. This PDF is not meant to be a 300-page textbook discussing nonsense that won't show up on USMLE. I tell you exactly what you need to know for the exam without the superfluous garbage.

I have a separate HY Neuroanatomy PDF where I go into detail regarding neuroanatomic structures showing cross-sections and labeled diagrams, etc. In this PDF, I do not do that. So if you need your hand held (i.e., "This is a pic of the brain. This is a pic of the spinal cord." Etc.), the other HY Neuroanatomy PDF will hit many of those points for you.

Stroke risk factors
- The two most important risk factors for stroke (cerebral infarction) on USMLE are hypertension and atrial fibrillation. - Regarding HTN, a strong systolic impulse pounds the carotid arteries, leading to endothelial damage and increased development of atherosclerotic plaques (carotid stenosis), which then launch off to the brain/eye. - Atrial fibrillation results in turbulence and stasis, leading to left atrial mural thrombus, which launches off. - Hypertension is most common in the population for causing stroke. Blood pressure control is more important than smoking cessation for decreasing stroke risk. - If the USMLE Q gives you a patient who has both AF and HTN, they want AF as the most important risk factor. In other words, even though HTN is the most common risk factor in the population for stroke, AF is still more likely to cause stroke if the patient has both. - There are other details regarding stroke I will discuss in this PDF, but for starters, these risk factors are pass-level points for understanding how/why most strokes occur – i.e., "How do most strokes occur?" → "Oh, well the patient will usually have carotid stenosis from HTN, where an atheromatous plaque has launched off, or will have AF, where an LA mural thrombus has launched off." Simple. - I discuss carotid stenosis and AF management for stroke prevention in detail in the HY Cardio PDF.

Circle of Willis HY strokes	
Affected vessel	**HY Points**
ACA	- Anterior cerebral artery. - Motor/sensory abnormalities of contralateral leg.
MCA	- Middle cerebral artery. - Motor/sensory abnormalities of contralateral arm + face. - Dominant MCA stroke (usually left) can lead to Wernicke or Broca aphasia (discussed below). - Non-dominant MCA stroke (usually right) can cause hemispatial neglect (inability to draw clockface).
PCA	- Posterior cerebral artery. - Contralateral homonymous hemianopsia with macular sparing. - Prosopagnosia (inability to recognize faces).

Other stroke syndromes	
Lateral medullary syndrome	- Aka Wallenberg syndrome. - The answer for **dysphagia + ipsilateral Horner syndrome** after a stroke. - Due to stroke of posterior inferior cerebellar artery (PICA) or vertebral artery. - PICAchew (Pikachu) → PICA stroke causes dysphagia. - Students might know Horner syndrome (ipsilateral miosis, partial ptosis, anhidrosis) can be caused by Pancoast tumor, but another HY cause is lateral medullary syndrome.
Medial medullary syndrome	- The answer for **ipsilateral tongue deviation** after a stroke. - Due to stroke of anterior spinal artery.
Lateral pontine syndrome	- The answer for **ipsilateral Bells palsy** after a stroke. - Due to stroke of anterior inferior cerebellar artery (AICA). - FACIAL (contains AICA backwards) → "That's the one that's Bells palsy."
Weber syndrome	- Midbrain stroke. - The answer for **ipsilateral CN III palsy** (i.e., down and out eye) + contralateral spastic hemiparesis (weakness).
Locked-in syndrome	- Basilar artery stroke. - The answer for **inability to move entire body** except for eyes.
Gerstmann syndrome	- Stroke of angular gyrus of parietal lobe.

	- Tetrad of 1) agraphia (inability to write); 2) acalculia (cannot do math); 3) finger agnosia (can't identify fingers); 4) left-right disassociation (cannot differentiate between left and right sides of body).
Hemiballismus	- Stroke of subthalamic nucleus. - Causes "ballistic" flailing of contralateral arm and/or leg. - Weird condition, but for some reason asked. For example, if they say patient has random flailing of left arm following a stroke, answer = right subthalamic nucleus.

Lenticulostriate strokes	
Lacunar infarcts	- HTN can cause lipohyalinosis (fancy word for microatheroma formation) of small lenticulostriate arteries deep within the brain. - Ischemia within these small vessels leads to necrosis and reabsorption of tissue, leading to tiny cavities called lacunae. - Unlike larger strokes, lacunar infarcts often do not present with cortical deficits like aphasia, neglect, or visual field losses. Instead, they manifest as specific syndromes based on their location, such as pure motor hemiparesis, pure sensory stroke, or ataxic hemiparesis. The USMLE doesn't give a fuck about the specific types of lacunar deficits/strokes. They just what you to know they exist / are possible, and that the mechanism is HTN causing lipohyalinosis of these small, penetrating vessels, as mentioned above.
Charcot-Bouchard microaneurysms	- Charcot-Bouchard microaneurysms are tiny (<1mm) aneurysms that form within lenticulostriate arteries that can bleed and cause hemorrhagic strokes and intraparenchymal (intracerebral) bleeds.

Neuro necrosis points (Step 1 only)
- Liquefactive necrosis is the answer for necrosis of nervous system tissue. - "Red neurons" is the answer for what will be seen acutely with ischemic infarction of the CNS. This refers to their strong eosinophilic (pink) staining with H&E. This is in contrast to myocardial infarction, where "no histologic changes" is the answer immediately after. - Microglia are the resident macrophages of the CNS. They phagocytose necrotic brain/spinal tissue. - Astrocytes are the glial cells (non-neuronal cells of CNS/PNS) that proliferate and become a glial scar (gliosis). In other words, if they ask which cell is responsible for scar formation in CNS, answer = astrocyte.

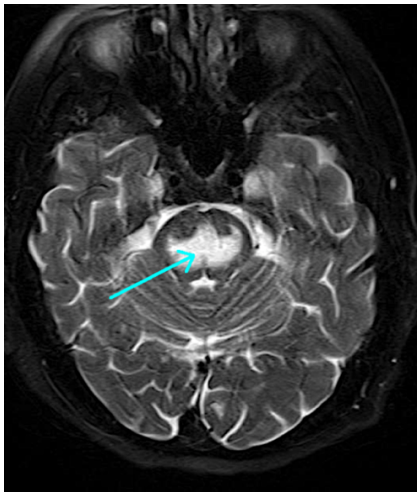
Wallerian degeneration (Step 1 only)
- Wallerian degeneration is a term that refers to degradation of an axon/myelin sheath distal to the site of injury. Regrowth occurs at maximum of 1mm/day. Both PNS and CNS neurons undergo Wallerian degeneration, but regeneration occurs within the PNS, not CNS. This is because PNS Schwann cells can regenerate myelin, whereas CNS oligodendrocytes do not effectively regenerate myelin post-injury. - The optic nerve is considered an extension of the CNS and is myelinated by oligodendrocytes; the other cranial nerves are part of the PNS and are myelinated by Schwann cells. This distinction is why degeneration of the optic nerve results in permanent blindness, whereas other facial nerves (e.g., CN VII for Bell's palsy) have better regenerative potential. The caveat, however, is that the extent of CN recovery depends on the nature of the injury and is not always complete, the same way nerve injuries in the limbs might not fully recover either.

Vertebrobasilar insufficiency	
Subclavian steal syndrome	- Confusing condition when you're first learning things. - The vertebral artery (goes to brain) is the first branch of the subclavian artery (goes to arm). - If there is a narrowing/stenosis of the proximal subclavian prior to the branch point of the vertebral artery, this can lead to lower pressure in the vertebral artery. - This can cause a backflow of blood in the vertebral artery, producing miscellaneous neuro findings such as dizziness. - **Blood pressure is different between the two arms.** - USMLE will ask the Q one of two ways: - 1) they'll give you dizziness in someone who has BP different between the arms and then ask for merely "subclavian steal syndrome," or "backflow in a vertebral artery" as the answer. - 2) they'll give you BP in one of the arms + give you dizziness, then the answer will be, "Check blood pressure in other arm." - Next best step in Dx is CT or MR angiography (asked on 2CK NBME). - I should point out that probably ¾ questions on USMLE where blood pressure is different between the arms, this refers to aortic dissection. But ¼ is subclavian steal syndrome. As per my observation.
Vertebral artery stenosis	- Presents same as subclavian steal syndrome with otherwise unexplained dizziness, but blood pressure is not different between the arms because the subclavian is not affected. - Caused by atherosclerosis. CT or MR angiography can diagnose. - "Vertebrobasilar insufficiency" is a broader term that refers to patients who have either subclavian steal syndrome or vertebral artery stenosis.
Vertebral artery dissection	- A false lumen created by dissection in a vertebral artery can lead to stasis and clot formation, which in turn can embolize to the brain and cause stroke. - NBME can mention recent visit to a chiropractor (neck manipulation is known cause). - The answer on the NBME is **heparin** for patients who have experienced posterior stroke due to vertebral artery dissection. Sounds weird because it's arterial, but it's what USMLE wants.

HY Aphasia types	
Wernicke	- Fluent aphasia (i.e., patient can speak with normal pace + use lots of words), but none of it makes sense ("word salad"). - Ability to comprehend is similarly impaired (i.e., patient makes no sense in terms of what comes out + inability to make any sense of what comes in). - Repetition is impaired (i.e., patient cannot repeat back a sentence spoken to him/her). - Wernicke area is located in the temporal lobe; caused by L-sided MCA infarct.
Broca	- Non-fluent aphasia (i.e., patient has "telegraphic speech"), where there is frustration in not being able to communicate despite comprehending normally, akin to trying to communicate in a second language. - Repetition impaired. - Broca area is located in frontal lobe; caused by L-sided MCA infarct.
Conductive	- Patient only has repetition impaired. - Stroke of arcuate fasciculus, which is bundle of nerve fibers that connects Wernicke and Broca areas.
Global	- Presentation akin to having Broca and Wernicke aphasias at the same time, in addition to repetition being impaired. - Stroke of Broca + Wernicke areas + arcuate fasciculus.
Transcortical sensory	- Presentation same as Wernicke but repetition intact.

Transcortical motor	- Presentation same as Broca, but repetition intact.

Stroke management / Tx	
Ischemic stroke	- Non-contrast CT of the head is done to look for ischemic vs hemorrhagic stroke. If blood is present, it will appear as bright (hyperdense) areas on CT. - Contrast CT, in contrast (no pun intended, LOL!), is done for stuff like brain abscess and malignancy. But for suspected bleeds, we do non-contrast. - If stroke is ischemic (i.e., no blood seen on non-contrast CT), tPA is given if symptom onset has been within the past 4.5 hours and there are no overt contraindications to tPA (discussed below). - The start of the 4.5-hour window refers to when the patient was last confirmed/observed to be normal. So if the patient was sleeping prior to the stroke, it must be known exactly when the patient started sleeping, and if he/she was normal prior to going to bed. - The USMLE can be tricky about this point, where they might say a guy awoke this morning from sleep 2 hours ago + has facial drooping, where tPA is wrong because the stroke could have occurred at any point while he was asleep + the last time he was observed to have no stroke signs was the night before. - If the 4.5-hour window has elapsed, aspirin is the answer for ischemic stroke. - BP should be brought down rapidly until below 185/110 so that tPA can be given. Below this threshold, cautious/slower BP control should be done, since ↑ BP can actually facilitate perfusion of penumbric areas of ischemia (i.e., areas of ischemia around the stroke).
Hemorrhagic stroke	- Do not give tPA. First step is lowering BP (labetalol, nicardipine, or hydralazine). - Reduce BP rapidly to systolic <140 mm Hg. - Reverse anticoagulation if the patient is on it (i.e., FFP for patients on warfarin).

Sodium correction	
Cerebral edema	- Correction of hypernatremia too quickly with hypotonic saline can cause cerebral edema.
Central pontine myelinolysis	- Correcting hyponatremia too quickly with hypertonic saline causes central pontine myelinolysis. - Can be described as “osmotic demyelination.” - Causes locked-in syndrome.  - USMLE can show MRI of the pons similar to the above.

Hypercalcemic crisis
- Refers to cognitive dysfunction / a delirium-like state in the setting of severe hypercalcemia, often due to malignancy or primary hyperparathyroidism. - USMLE wants you to know that high calcium, as well as any sodium disturbance, can cause delirium. - First step on USMLE for Tx of hypercalcemia is normal saline. - After normal saline, USMLE wants bisphosphonate therapy (I've seen pamidronate listed on NBME). - I've never seen calcitonin or loop diuretics as correct answers for hypercalcemia Tx.

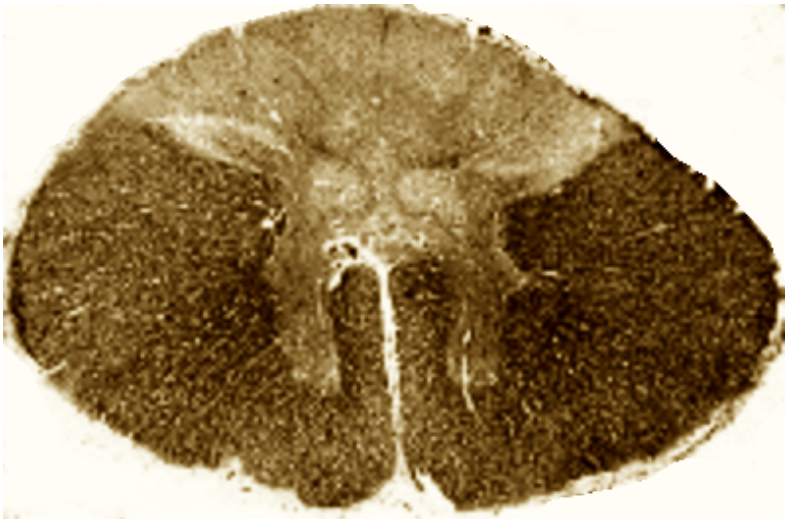
Pseudotumor cerebri
- Term that refers to increased intracranial pressure in the setting of no clear structural cause. - Q can mention papilledema (optic disc swelling) or absent venous pulsations on fundoscopy (another increased ICP buzzy finding) in any of the following scenarios/drugs: - Obesity (idiopathic intracranial hypertension), OCPs, hypervitaminosis-A, isotretinoin, and danazol.

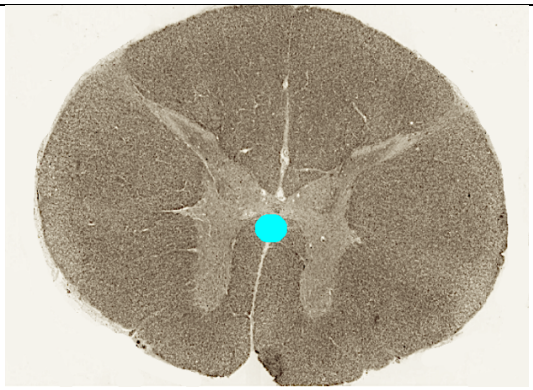
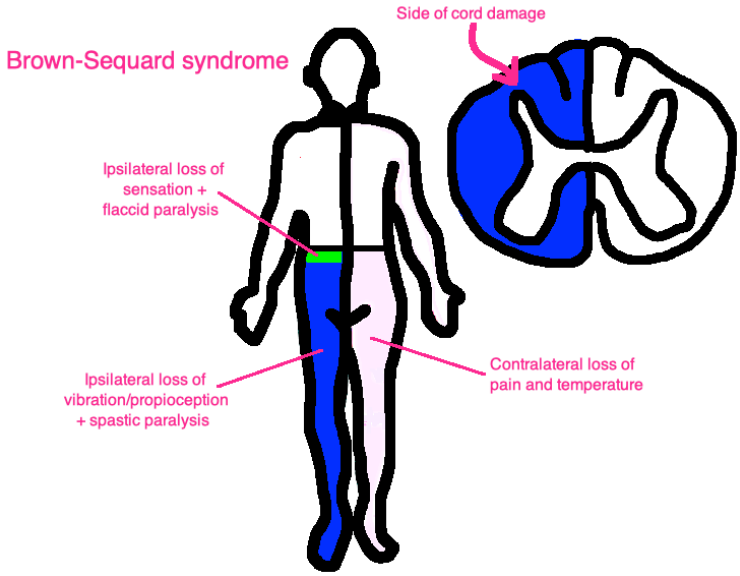
Reye syndrome
- Cerebral edema and hepatotoxicity seen with administration of aspirin to children (and sometimes adolescents) during a viral infection. - Thought to be due to impairment of beta-oxidation, but USMLE doesn't care. - Don't confuse with anticholinergic delirium, which can present as confusion with meds such as diphenhydramine (a 1st-generation H1 blocker).

Diffuse axonal injury
- Deceleration injury (e.g., car accident) followed by severe cognitive and/or motor/sensory dysfunction (i.e., following an MVA, the patient has to relearn tying his/her shoes, etc.). - USMLE can show MRI showing scattered hyperintense (white) lesions. 

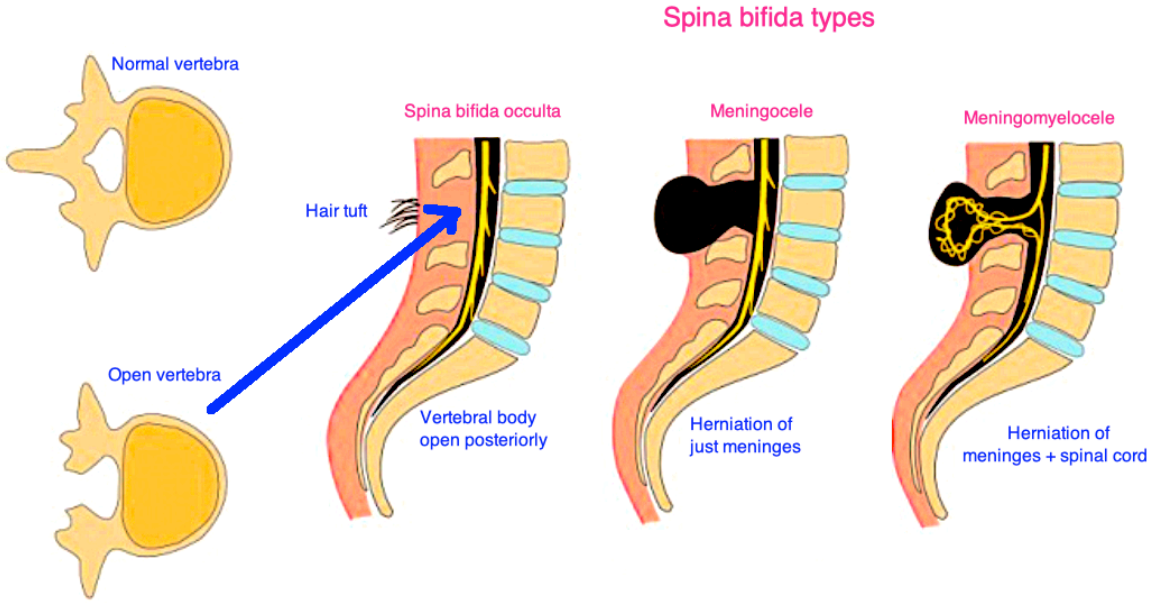
Pass-level spinal tract functions	
Spinothalamic tract	- **Pain and temperature sensation from contralateral body.** - Cross-over (decussation) point is in spinal cord, meaning if, e.g., the left spinothalamic tract in the spinal cord is fucked up, we lose pain and temperature on the right side of the body below the lesion. - Decussation point is called anterior white commissure. A lesion at this point is called syringomyelia, where there is bilateral loss of pain and temperature below the lesion.
Corticospinal tract	- **Motor function from ipsilateral body.** - In other words, if left-sided corticospinal tract is fucked up, we lose motor function on the left side below the lesion.
Dorsal columns	- **Carry vibration + proprioception from ipsilateral body.** - Tabes dorsalis is dorsal column lesion caused by neurosyphilis. - If the dorsal columns are disrupted, patient will have (+) Romberg test, which is where he/she falls over when standing with eyes closed. I discuss this more below.

Upper- vs lower motor neuron findings	
UMN	- Lesion of the corticospinal tract (i.e., lesion from the cerebral cortex down until the anterior horn in the spinal cord). - Present as hyperreflexia, hypertonia**,** clonus, Babinski sign.
LMN	- Lesion involving the motor neuron from the anterior horn until the muscle itself. If the USMLE gives you an anterior horn lesion, this will be LMN, not UMN. - Present as hyporeflexia, hypotonia/flaccidity, fasciculations.

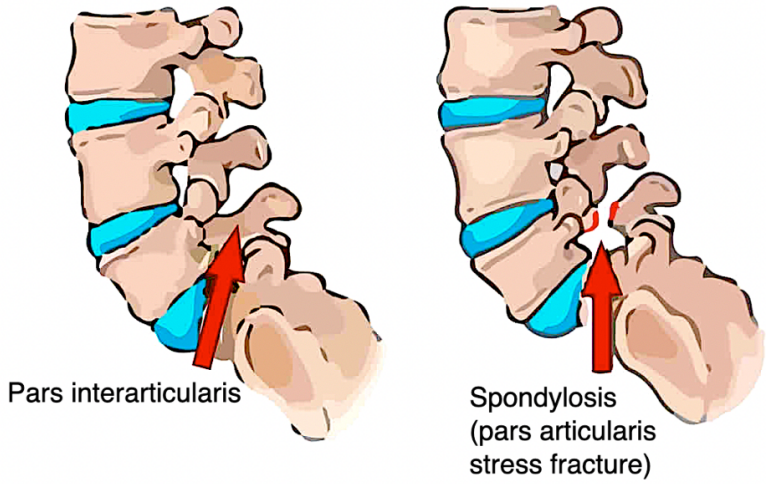
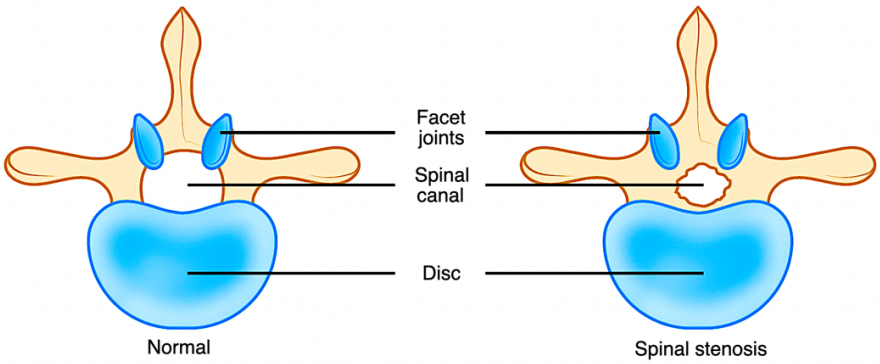
HY Spinal conditions	
Tabes dorsalis	- Loss of dorsal columns due to neurosyphilis.  Myelin stain used, where normal myelination appears black. The dorsal columns are fucked up here, which is why they're white. - Bilateral loss of vibration + proprioception. - Ataxia with (+) Romberg test → indicates dorsal columns are fucked up as the cause of the ataxia, since proprioception is lost. In contrast, if the cause of ataxia is cerebellar, Romberg test is (-).
Syringomyelia	- 9/10 Qs will give bilateral loss of pain and temperature due to lesion of anterior white commissure (decussation point for spinothalamic tract).

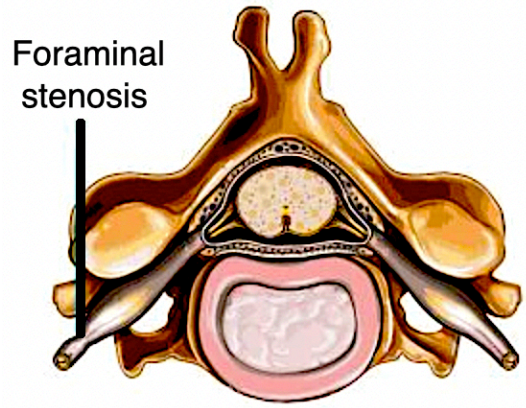
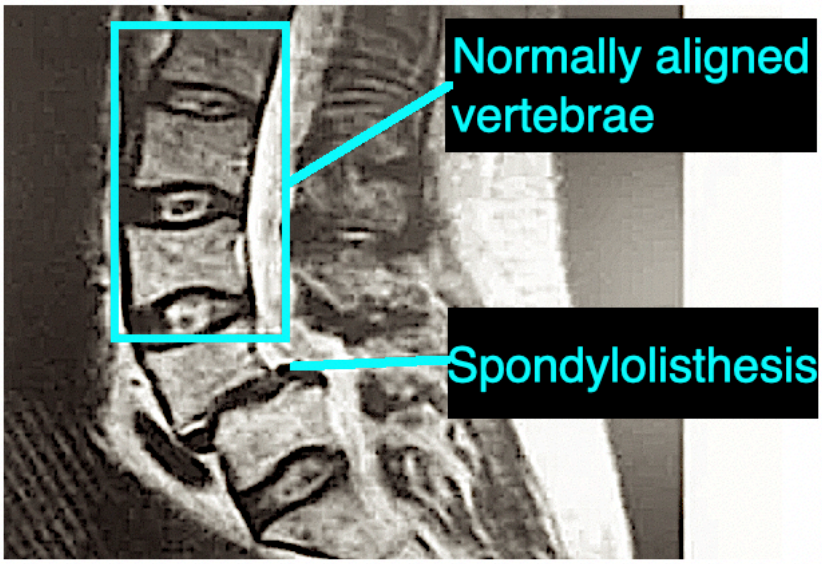
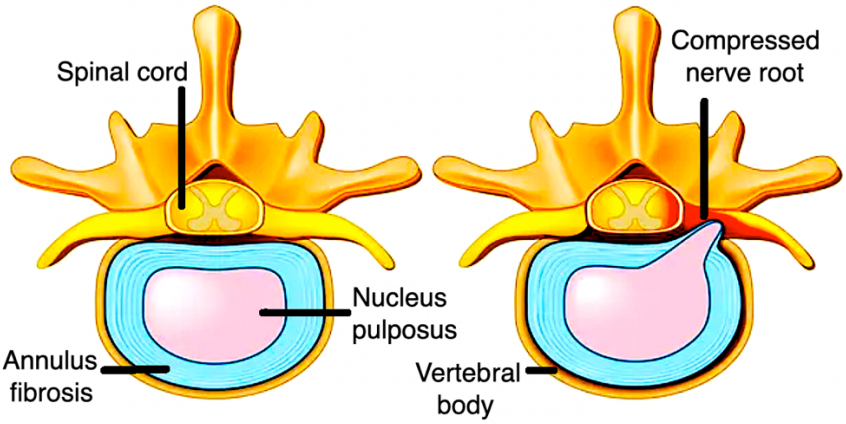
	 Cyan dot marks the anterior white commissure. - Maybe 1/3 of Qs will also involve corticospinal tract, where you'll see some impaired motor function in addition to the bilateral loss of pain/temperature. - 1/10 Qs will not mention pain/temperature loss but instead just give impaired motor function, where it sounds nothing like syringomyelia, but you're forced to eliminate to get there (on a 2CK Neuro CMS form).
Brown-Sequard syndrome	- Hemisection of spinal cord.  - Ipsilateral loss of vibration + proprioception (dorsal columns). - Ipsilateral loss of motor function (corticospinal tract). - Contralateral loss of pain + temperature (spinothalamic). - Not typically caused by stab wound where half the spinal cord is perfectly transected. It's to my observation on NBME forms that they like viral infection or SLE causing transverse myelitis.
Transverse myelitis	- Inflammation of spinal cord resulting in variable deficits. - As mentioned above, USMLE likes viral infection and SLE (autoimmune). - Brown-Sequard syndrome is one way transverse myelitis presents.
Central cord syndrome	- The answer when weakness in the upper limbs is greater than weakness in the lower limbs. - Typically follows hyperextension injury of the neck during whiplash in a rear-end MVA.
Anterior cord syndrome	- The answer when there's preservation of vibration/proprioception but loss of motor + pain/temperature, classically in patient with arterial disease. - In other words, everything but the dorsal columns is fucked up. Essentially the opposite of tabes dorsalis.

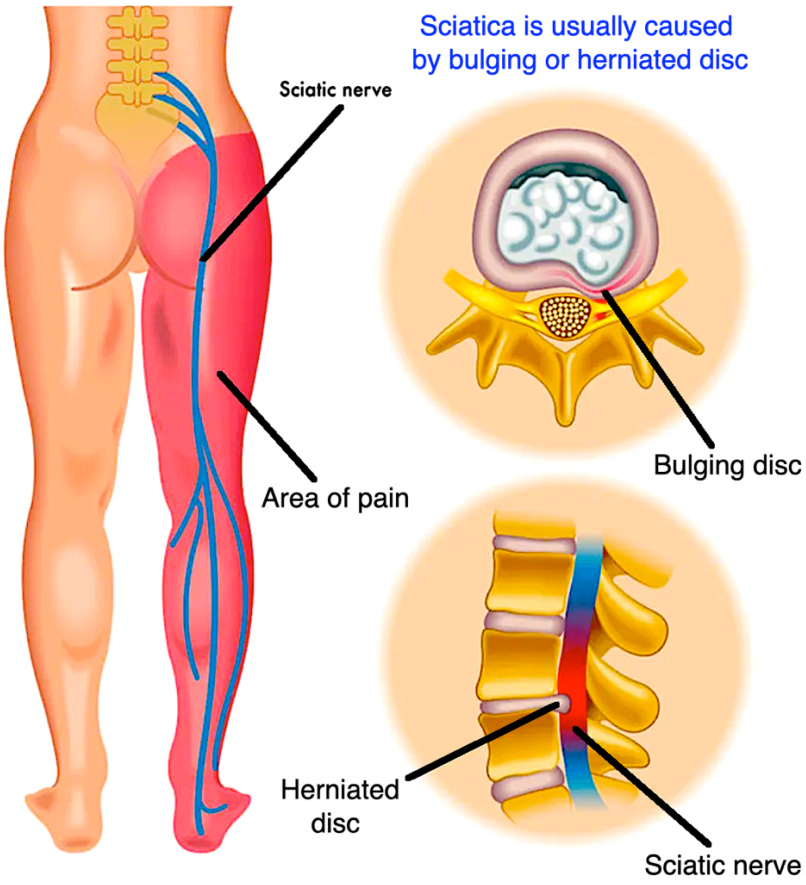
<table>
<tr><th colspan="2">Brachial plexus injuries</th></tr>
<tr><td colspan="2">- Both can be caused in neonates by breech / traumatic labor.
- Can also be caused by injuries such as grabbing onto tree branch while falling.</td></tr>
<tr><td>Klumpke</td><td>- Lower brachial plexus injury (C8-T1).
- Sometimes rather than Klumpke written as the diagnosis, the answer will just be “lower brachial plexus” when they ask for what’s injured.
 
Klumpke
“claw-hand”</td></tr>
<tr><td>Erb-Duchenne</td><td>- Upper brachial plexus injury (C5-C6).
- Sometimes rather than Erb-Duchenne written as the diagnosis, the answer will just be “upper brachial plexus” when they ask for what’s injured.
 
“Waiter’s tip”
deformity
- Arm is adducted, pronated, and wrist flexed.</td></tr>
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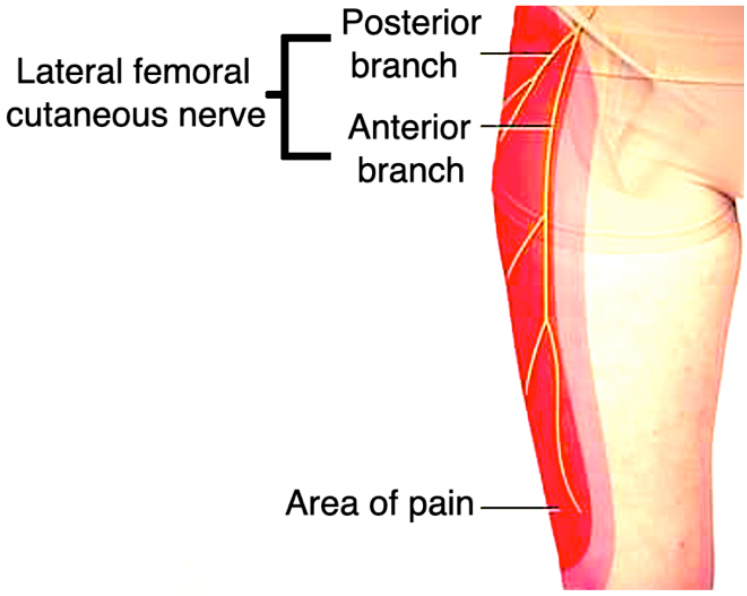
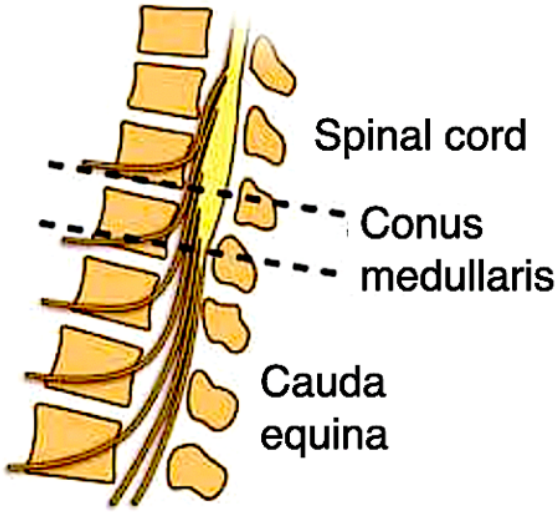
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<tr><th>Spina bifida</th></tr>
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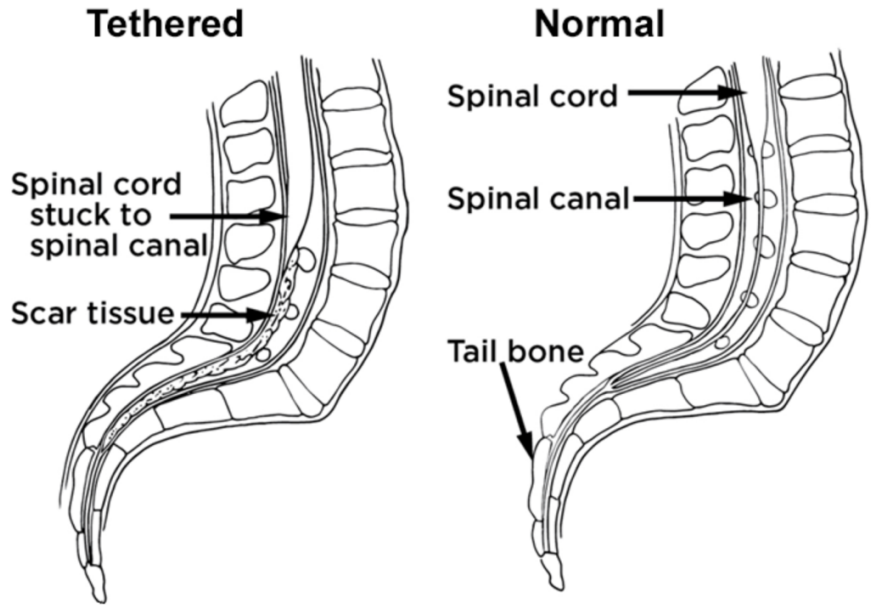
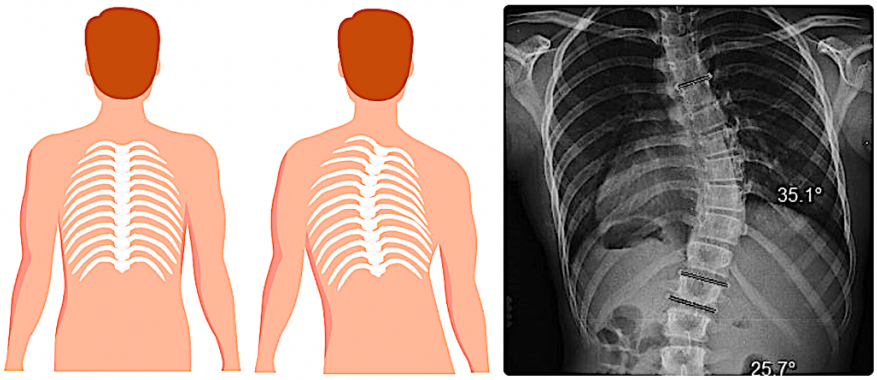
- Meningocele: Neural tube defect with herniation of just the meninges through the vertebral arch.
- Meningomyelocele: Neural tube defect with herniation of the meninges + spinal cord through the vertebral arch.
- HY point is that neural tube defects (spina bifida) occur between 3-4 weeks gestation.
- The pregnant woman needs adequate folic acid (B9) prior to and during early pregnancy for prevention.
- Antiepileptics (i.e., particularly valproic acid, but also phenytoin and carbamazepine) are also associated with neural tube defects.
- Neural tube defects lead to increased AFP and acetylcholinesterase in CSF.</td></tr>
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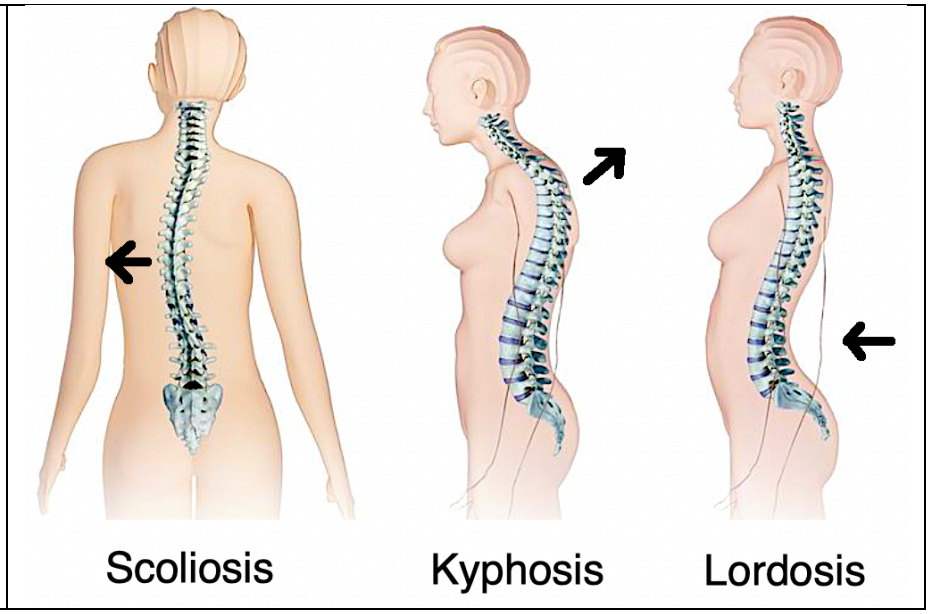
HY spinal conditions for USMLE	
Cervical spondylosis	- The answer on USMLE if they say patient over 50 has neck pain + MRI shows degenerative changes of cervical spine. - Can occur in lumbar spine, but USMLE likes cervical spine for this. - Technically defined as degeneration of pars interarticularis component of vertebral body. 
Atlantoaxial subluxation	- Increased mobility between the first (atlas) and second (axis) vertebrae. - Really HY in patients who have advanced **rheumatoid arthritis.** - Must do CT or flexion/extension x-rays of cervical spine prior to surgery when a patient will be intubated; I've seen both of these as answers for NBME Qs. - One Q gives patient with RA not undergoing surgery who has paresthesias of upper limbs → answer is just MRI of cervical spine (implying atlantoaxial subluxation has already occurred).
Lumbar spinal stenosis	- Narrowing of the spinal canal. Presents two ways on USMLE: - 1) The answer if they mention a patient over 50 who has lower back pain that's worse when walking down a hill (i.e., relieved when leaning forward), or when standing/walking for extended periods of time. - 2) Can cause "neurogenic claudication," where the vignette sounds like the patient has intermittent claudication, but they'll make it clear the peripheral pulses are normal and that the patient doesn't have cardiovascular disease. - Technically an osteoarthritic change of the spine; therefore increased risk in obesity (but not mandatory for questions). 
Cervical foraminal stenosis	- The answer on USMLE if they say old woman has difficulty fastening buttons + weakness of hand muscles + loss of sensation of little finger → answer = "C7-T1 foraminal stenosis." - Not stenosis of cervical spinal canal, but stenosis of foramen where nerve exits.

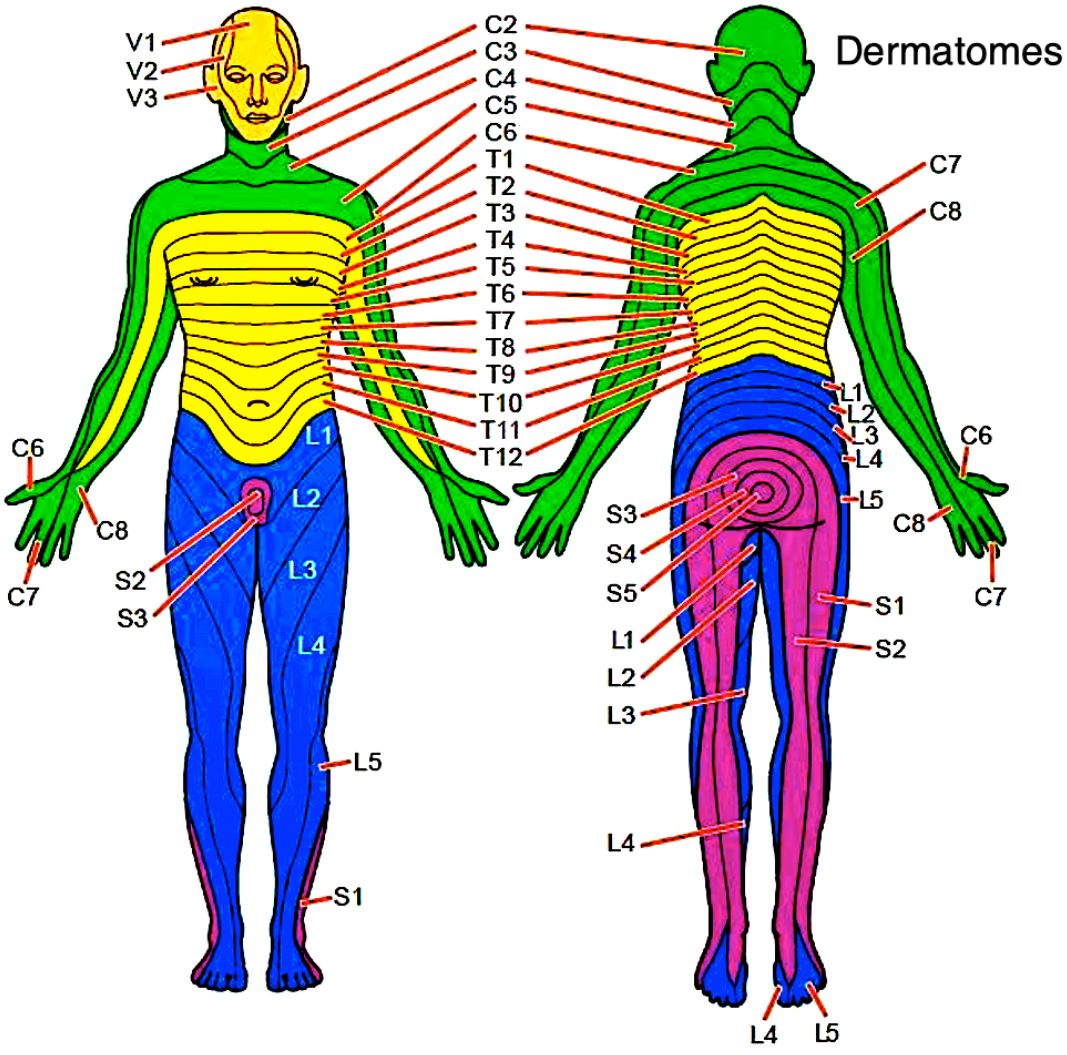
	
Spondylolisthesis	- Answer on USMLE if they say there's a "step-off" between infra- / suprajacent vertebrae. In other words, they'll say one vertebra "juts out" or has a "step-off" compared to those above/below it. - Can be due to trauma or idiopathic development. 
Disc herniation	- Herniation of nucleus pulposus through a tear in annulus fibrosis.  - The answer on USMLE if they mention **radiculopathy** (i.e., shooting pain down a leg) after lifting a heavy weight or bending over (e.g., while gardening). They can write the answer as "herniated nucleus pulposus" or "tear in annulus fibrosis." - Be aware that cervical disc herniation "is a thing," meaning it's possible and also assessed on USMLE. The NBME asks this a couple times, where patient has shooting pain down an arm, and answer is "C8 disc herniation."

	- If suspected, the newest NBMEs want "no diagnostic studies indicated." X-ray and MRI are not indicated unless there is motor/sensory abnormality (i.e., weakness or numbness). But for mere radiculopathy (i.e., radiating pain), no imaging is necessary on new NBME content. - Straight-leg raise test eliciting pain suggests disc herniation. The Q will say there is pain with straight-leg test → next best step = "no further diagnostic studies indicated." - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Lumbosacral strain	- The answer on USMLE if they say patient has paraspinal muscle spasm following lifting of heavy box **without** radiculopathy. If they say radiculopathy, the answer is disc herniation instead. - Straight-leg test can cause pain (i.e., **negative** test). The test is only positive if they say it reproduces radiating pain. - Do **not** x-ray. This is really HY for USMLE. Apparently lumbar spinal x-rays are one of the most frivolously ordered tests, and USMLE wants you to know that you do not order one for simple lumbosacral strain (or suspected disc herniations). - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Sciatica	- 90% of the time is due to disc herniation.  - Straight-leg test classically (+) – i.e., reproduces radiating pain. - Tx = Light exercise as tolerated + NSAIDs. Bed rest is wrong answer on USMLE. - On one of the NBMEs, ibuprofen straight-up is listed as the answer.
Meralgia paresthetica	- The answer on USMLE if they say patient has pain or paresthesias running down the lateral thigh. - Due to entrapment of lateral femoral cutaneous nerve.

	
Conus medullaris syndrome	- Refers to injury/lesion at the terminal part of the spinal cord, around T12 to L2. - Acute-onset + bilateral leg weakness and/or pain. - Urinary retention. - Classically perianal anesthesia. - Caused by trauma, tumors, vascular events. 
Cauda equina syndrome	- Refers to injury or lesion affecting the bundle of nerve roots below the conus medullaris – i.e., the cauda equina ("horse's tail"). - Usually gradual-onset + asymmetric weakness and/or pain. - Urinary retention. - Classically saddle anesthesia. - USMLE likes disc herniation and metastases to the spine as highest-yield causes. - For example, the Q can give 55-yr-old man with lower-limb neurologic findings + urinary retention, and the answer = "metastases to cauda equina."

Tethered cord	 - Caused by abnormal attachment of the spinal cord to the wall of the spinal canal, usually by scar tissue. - This attachment results in abnormal stretching of the spinal cord and variable neurologic deficits. - Caused by myelomeningocele in pediatrics, or by trauma/surgery in adults. - Diagnosis of exclusion on USMLE (meaning we eliminate to get there).
Scoliosis	- Sideways curvature of spine, creating an S- or C-shaped curve. - Usually idiopathic; affects 3% of population; girls 4:1.  Normal Spine Scoliosis Rib Rotation Scoliosis X-Ray with Rib Rotation - Can be associated with Marfan syndrome, Friedreich ataxia, NF1. - Adams forward bend test used to diagnose. - USMLE wants you to know most children do not need treatment, but that curvatures will remain throughout life. - If curvature is >25 degrees and child is still growing, bracing can be used. - I've never seen surgery as answer for scoliosis on NBME; literature says recommended only when curvature >40 degrees.
Kyphosis	- Abnormal convex curvature of thoracic spine. - Usually idiopathic due to old age; can be due to degenerative disc disease and compression fractures (osteoporosis). - If severe, can in theory cause restrictive lung disease due to impaired chest wall expansion.


Scoliosis
Kyphosis
Lordosis


Dermatomes
V1
V2
V3
C2
C3
C4
C5
C6
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
C7
C8
L1
L2
L3
L4
L5
S1
S2
S3
S4
S5

Lower limb reflexes / radiculopathies	
L4 radiculopathy	- The answer on USMLE if patient loses knee (patellar) reflex + has weakened knee extension. - Pain / paresthesias / numbness in L4 distribution (anterior thigh + medial leg). - Disc herniation of L3-4. - Just remember that L4 is the one where the knee reflex is fucked up.
L5 radiculopathy	- The answer on USMLE if patient loses dorsiflexion. - Pain / paresthesias / numbness in L5 distribution (lateral + anterior leg). - Disc herniation of L4-L5.
S1 radiculopathy	- The answer on USMLE if patient loses ankle (Achilles) reflex + has weakened plantar flexion. - Pain / paresthesias / numbness in S1 distribution (sole of foot + lower leg). - Disc herniation of L5-S1. - Just remember that S1 is the one where the ankle reflex is fucked up. - **SALT** → **S**1, **A**chilles, **L**ateral leg dermatome, **T**ibial motor issue (plantar flexion).

Lower limb nerve HY Points for USMLE	
Common peroneal (fibular) nerve	- The answer on USMLE if patient loses both eversion **and** dorsiflexion of the foot. - Sensation to upper third of lateral leg (around and below lateral knee). - Splits into superficial and deep peroneal (fibular) nerves.
Superficial peroneal nerve	- The answer on USMLE if patient loses only eversion of the foot, but dorsiflexion stays intact. - Sensation to lower lateral leg and dorsum of foot.
Deep peroneal nerve	- The answer on USMLE if patient only loses dorsiflexion of the foot, but eversion stays intact. - **D**eep for **D**orsiflexion, which means superficial is the one that does eversion instead. - Loss of dorsiflexion causes a high-steppage gait (patient has to lift foot high into the air with each step). - Also does sensation to webbing between 1st and 2nd toes. I've never seen NBME Qs ask or give a fuck about this sensation detail, but students get fanatical about it as if it's supposed to be high-yield.
Tibial nerve	- The answer on USMLE if patient loses plantarflexion (can't stand on tippytoes) and/or inversion. - Sensation to bottom of foot / heel.
Sciatic nerve	- The answer on USMLE if patient has motor dysfunction of tibial and common peroneal nerves at the same time, or has sciatica (shooting pain down leg). - Splits into the common peroneal nerve and tibial nerve. - Does not supply sensation to thigh; sensation by the sciatic nerve encompasses that which is supplied by the combination of the common peroneal nerve and tibial nerves. - Supplies some motor function to muscles of thigh but USMLE doesn't care. - Sciatica = shooting pain from the lower back down the leg usually as the result of disc herniation; NBME Q has NSAIDs as treatment; straight-leg test is classically used in part to diagnose, but I've seen this test show up on NBME material for simple lumbosacral strain (i.e., the test is non-specific and not reliable).
Obturator nerve	- The answer on USMLE if patient has inability to adduct the hip, with loss of sensation to medial thigh.
Femoral nerve	- The answer on USMLE if patient cannot extend and/or has buckling at the knee. - Also does sensation to anterior thigh + medial leg (not thigh), although I haven't seen sensation specifically asked for femoral nerve.
Saphenous nerve	- The answer on USMLE if patient loses sensation to medial calf. - Pure sensory branch of the femoral nerve.

Sural nerve	- The answer on USMLE if patient loses sensation to lower lateral calf. In contrast, if sensation loss is upper lateral leg, that's common peroneal nerve instead. - Often confused with saphenous. Good way to remember is: sura**L** is **L**ateral, therefore saphenous must be the one that's medial. - NBME has Q where they ask which nerve could be used for a nerve graft. The answer is sural nerve. This is not about specifically memorizing sural nerve here. The Q gives 4 motor nerves + only one sensory nerve. So if we want to harvest a nerve for a graft, it's better to compromise sensory function over motor function.
Superior gluteal nerve	- The answer on USMLE if patient has Trendelenburg gait → opposite side of pelvis will fall while walking, so patient will tilt trunk toward side of lesion while walking to maintain level pelvis. - Innervates gluteus medius and minimus.
Inferior gluteal nerve	- The answer on USMLE if patient cannot squat, stand up from a chair, or go up/down stairs. - Innervates gluteus maximus.

Neurogenic joint
- Aka Charcot joint, where patient injures joint due to lack of joint sensation from peripheral neuropathy. - Usually seen in diabetes; can also be seen in neurosyphilis. - The answer on USMLE when they say diabetic patient has "disorganization of the tarsometatarsal joints" on foot x-ray.

Upper limb nerve HY Points for USMLE	
Axillary	- Main innervation of the deltoid, allowing for abduction of arm 15-90 degrees. USMLE wants you to know deltoid has an origin on the lateral clavicle and axillary nerve innervating it is at C5/C6. - Palsy caused by **surgical neck of humerus fracture.** - USMLE vignette will often say "flattened deltoid" or "loss of sensation over lateral upper arm / deltoid."
Median	- Main innervation is lateral "3 and a half" fingers / thenar pad, and lateral forearm. - Does thumb abduction/opposition (touching thumb to pinky). This is in contrast to thumb adduction (touching thumb to index finger), which is ulnar nerve. - NBME wants "palmar cutaneous branch of median nerve" as answer for sensation over thenar region. - Palsy caused by **supracondylar fracture of humerus**, or "**distal shaft fracture."** Former is buzzy; latter sounds non-specific, but I've seen it this way on NBME. - Entrapment of median nerve causes carpal tunnel syndrome; will present as paresthesia/numbness of lateral hand / thenar region; can be caused by hypothyroidism (GAG deposition), acromegaly (growth of tendons), and pregnancy (edema); can occur bilaterally in construction workers using jackhammer. - Tx for carpal tunnel ultra-HY on 2CK FM forms. "Use of wrist pad when using computer" is answer on new 2CK NBME. If not listed, "wrist splint" is answer on FM form. If vignette says wrist splint fails, NSAIDs are wrong answer and not proven. USMLE wants "triamcinolone injection into carpal tunnel" (not IV steroids) as next answer. Surgery is always wrong answer for carpal tunnel on USMLE. - 2CK wants "electrophysiological testing" and "electromyography and nerve conduction studies" as next best step in diagnosis for carpal tunnel. There are NBME Qs where they have these as answers, where Phalen test / Tinel signs are not listed.
Ulnar	- Main innervation of medial "1 and a half" fingers, and medial forearm. - Ulnar nerve also does **finger abduction and adduction** (i.e., interosseous muscles).

	- USMLE loves **Froment sign** for ulnar nerve injury, which is inability to pinch a piece of paper between the thumb and index finger without bending in the index finger (ulnar nerve needed for thumb adduction against index finger, despite thumb being most lateral digit). - Distal compression (i.e., of wrist and hand only, not forearm) is aka **Guyon canal syndrome** and is caused by **hook of hamate fracture;** this can sometimes be seen in cyclists due to handlebar compression; presents as paresthesias / numbness of 4th and 5th fingers + hypothenar eminence. - Proximal compression (i.e., medial forearm + wrist/hand) is aka **cubital tunnel syndrome** and is one of the most underrated diagnoses on USMLE, since its yieldness, especially on 2CK, is comparable to carpal tunnel syndrome, but students often haven't heard of it. Essentially, patient will get paresthesias of medial forearm + hand, where it "sounds like carpal tunnel but on the ulnar side instead" → answer = cubital tunnel syndrome. - Tx for cubital tunnel syndrome is "overnight elbow splint." Surgery is wrong answer on USMLE.
Radial	- Main innervation for finger, wrist, and elbow extension. - Innervates **BEST** → **B**rachioradialis, **E**xtensors, **S**upinator, **T**riceps. - Palsy occurs as a result of **midshaft fracture of the humerus,** or as a result of fracture at the **radial groove** (latter is obvious). - Retired Step 1 NBME Q says construction worker sustains "comminuted spiral fracture of humerus" (unusual, since spiral fracture classically = child abuse), and they ask for the resulting defect → answer = "loss of radial nerve function." - Highest yield point is that injury results in pronated forearm + wrist drop.
Musculocutaneous	- Main innervation of the biceps. - Just need to know injury results in loss of sensation over lateral forearm + decreased biceps function. - USMLE doesn't give a fuck about what kind of injury causes palsy.

Mononeuritis multiplex	
- Weird neuropathy that means neuropathy of "one large nerve in many locations" – e.g., wrist drop + foot drop in same patient. - Presents in patients with one of the vasculitides (i.e., Wegener, Churg-Strauss, microscopic polyangiitis). - In other words, you'll get a vignette of Wegener, and then you're like, "Wait, why does he/she have wrist drop though?" That's just mononeuritis multiplex.	
Granulomatosis with polyangiitis	- Formerly known as Wegener granulomatosis. - Answer on USMLE for adult with triad of 1) hematuria, 2) hemoptysis, and 3) "head-itis" – i.e., any problem with the head, such as nasal septal perforation, mastoiditis, sinusitis, otitis. - Associated with cANCA and anti-proteinase 3 (anti-PR3) antibodies. - Causes "necrotizing glomerulonephritis" that can lead to rapidly progressive glomerulonephritis (RPGN).
Eosinophilic granulomatosis with polyangiitis	- Formerly known as Churg-Strauss. - Presents as combo of asthma + eosinophilia +/- head-itis. - Head-itis always seen in Wegener vignettes, but maybe only ~50% of CS Qs. - Renal involvement rare for CS. - Associated with pANCA and anti-myeloperoxidase (anti-MPO) antibodies.
Microscopic polyangiitis	- Will just present as hematuria in a patient who is pANCA / anti-MPO (+). - Similar to Wegener, can cause RPGN.

Hydrocephalus	
Communicating	- CSF can flow freely between the ventricles and subarachnoid space around the brain and spinal cord. - The answer for meningitis causing hydrocephalus, where the mechanism is "failure of CSF reabsorption by the arachnoid granulations."
Non-communicating	- Aka obstructive hydrocephalus. - When there is a blockage preventing flow of CSF between the ventricles or subarachnoid space. - There is an NBME Q where they give a tumor of the spinal cord in the setting of hydrocephalus, and the answer is obstructive hydrocephalus. - Can be caused by aqueductal stenosis (narrowing of the cerebral aqueduct of Sylvius between the 3rd and 4th ventricles), which is the most common cause of congenital hydrocephalus. - Can be caused by colloid cyst, which is a benign fluid-filled growth at the anterior roof of the third ventricle, just posterior to the foramen of Monro.
NPH	- Normal pressure hydrocephalus. - "Wet, wobbly, wacky" +/- Parkinsonism. - In other words: urinary incontinence, ataxia, and cognitive dysfunction. - Due to impingement on the corona radiata (asked on an NBME), which is a white-matter bundle that connects the motor cortex superiorly to the internal capsule inferiorly. - Mechanism for urinary incontinence = "failure to inhibit the voiding reflex."
Congenital	- Aqueductal stenosis is most common cause. - Most important cause on NBME, in my opinion however, is congenital toxoplasmosis, which presents as a triad of 1) hydrocephalus, 2) chorioretinitis, and 3) intracranial calcifications.
Ex-vacuo	- Appearance of enlarged lateral ventricles on imaging without true hydrocephalus. - Can be caused by loss of surrounding brain matter. - Alzheimer and schizophrenia are important causes on USMLE, where enlargement of the lateral ventricles is a finding that can sometimes show up in question stems.

HY dementia types	
Alzheimer	- Gradual-onset idiopathic cognitive decline. - Patient usually have normal neurologic exam (i.e., no motor or sensory abnormalities) on USMLE. - MMSE score will be low (i.e., low-20s out of 30) on USMLE. If the diagnosis is instead benign senility, USMLE will give MMSE usually 28+. - Beta-amyloid plaques and neurofibrillary tangles (hyperphosphorylated tau protein) seen on brain biopsy. - Early-onset Alzheimer in Down syndrome (amyloid precursor protein gene is located on chromosome 21). - Presenilin gene mutations can cause Alzheimer (on NBME exam). Presenilin is a protein involved in the cleavage of amyloid. - Tx = cholinesterase inhibitors (donepezil, galantamine, rivastigmine); memantine (NMDA glutamate receptor antagonist) can also be used. - Sundowning is worsening of dementia at night that can resemble delirium. - 1st-line Tx for sundowning on NBME is "decrease ambient noise and distractions." "Bright illumination of the room at all times" is wrong answer.
Frontotemporal dementia	- Aka Pick disease. - Presents usually as triad of 1) personality change, 2) apathy, and 3) disinhibition. - Accumulation of hyperphosphorylated tau protein (similar to Alzheimer), except rather than accumulating as neurofibrillary tangles, it accumulates as round, silver-staining inclusions knowns as Pick bodies.
Lewy body dementia	- Dementia + visual hallucinations + Parkinsonism.

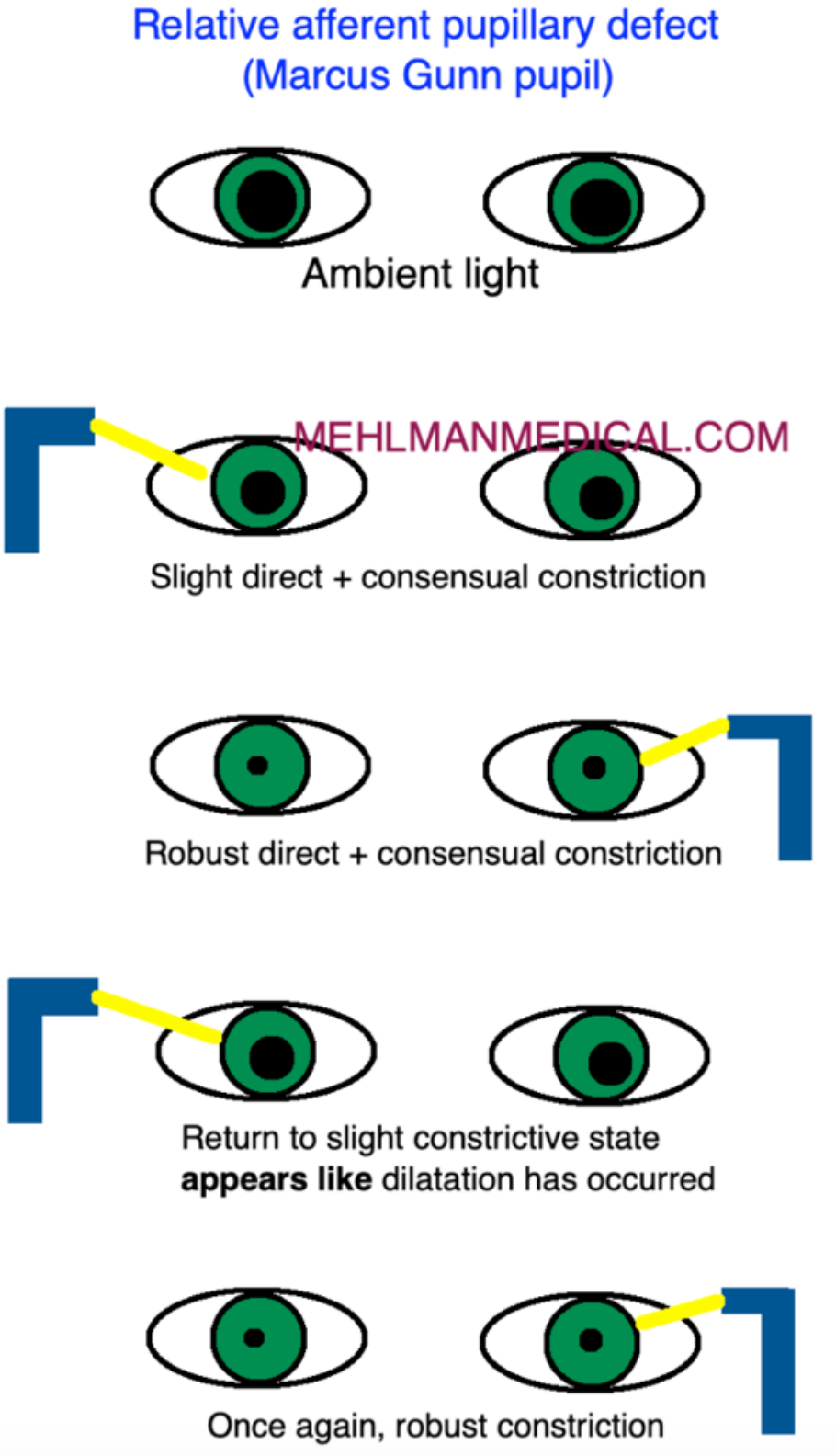
	- Lewy bodies are collections of alpha-synuclein. This protein is deposited throughout the brain in Lewy-body dementia. In Parkinson disease, in contrast, it is deposited primarily in the substantia nigra pars compacta of the midbrain.
Vascular dementia	- Aka multi-infarct dementia. - The answer for dementia + motor/sensory abnormalities. - Seen in patients who have repeated mini-strokes (cerebral infracts) due to hypertension. - Resources tend to focus on this notion of "step-wise decline" (i.e., concrete timepoints at which deficits started), but it's to my observation on NBME exams that this is rarely a salient aspect of Qs. What USMLE likes is giving motor and/or sensory deficits – i.e., you'll get a big paragraph with dementia, and you'll notice somewhere in the stem that the patient has, e.g., 3/5 strength in the right upper extremity. This indicates Hx of stroke.
AIDS	- Just be aware AIDS can cause dementia, known as AIDS complex dementia. - Can present as "wet, wobbly, wacky," similar to normal pressure hydrocephalus.
Pseudodementia	- Not actual dementia. This is depression that presents as cognitive decline. - Patients with depression who have apathy will perform poorly on the MMSE. - The Q might say the patient is unable to draw a clockface, but when prompted, is able to finish it quickly. They might also say patient remembers 0 out of 3 objects after 5 minutes. - Look for obvious signs of depression, such as short, quiet answers, and low mood.
Subacute combined degeneration	- The fancy name for neurologic degeneration seen in B12 deficiency. - Can present sometimes as a reversible cause of dementia. In elderly patients on tea and toast diets, or those in high-risk groups (i.e., vegans, pernicious anemia), B12 must be considered as cause of cognitive decline. - The patient can have peripheral neuropathy as a result of deficits to the 1) corticospinal tracts, 2) dorsal columns, and 3) spinocerebellar tracts. - The easy way to remember those three is to start by saying, "The spinothalamic tract is not involved." Then you say, "Well what are other ones I can think of?"
Hypothyroidism	- Hypothyroidism can also cause cognitive decline (e.g., in Hashimoto). - Checking TSH should be part of a patient's dementia workup. But be careful choosing this answer over obvious ones first, like simple depression screening for pseudodementia.
Neurosyphilis	- Just be aware that neurosyphilis is a reversible cause of dementia and should be considered. There's an NBME Q floating around for 2CK where they give (+) VDRL in 82-year-old woman with cognitive decline, and the treatment is penicillin.
Neurolyme	- Lyme disease can cause cognitive decline. - If the patient lives in endemic area + has risk factors like pet dogs or wilderness exposure, this should be considered as DDx.

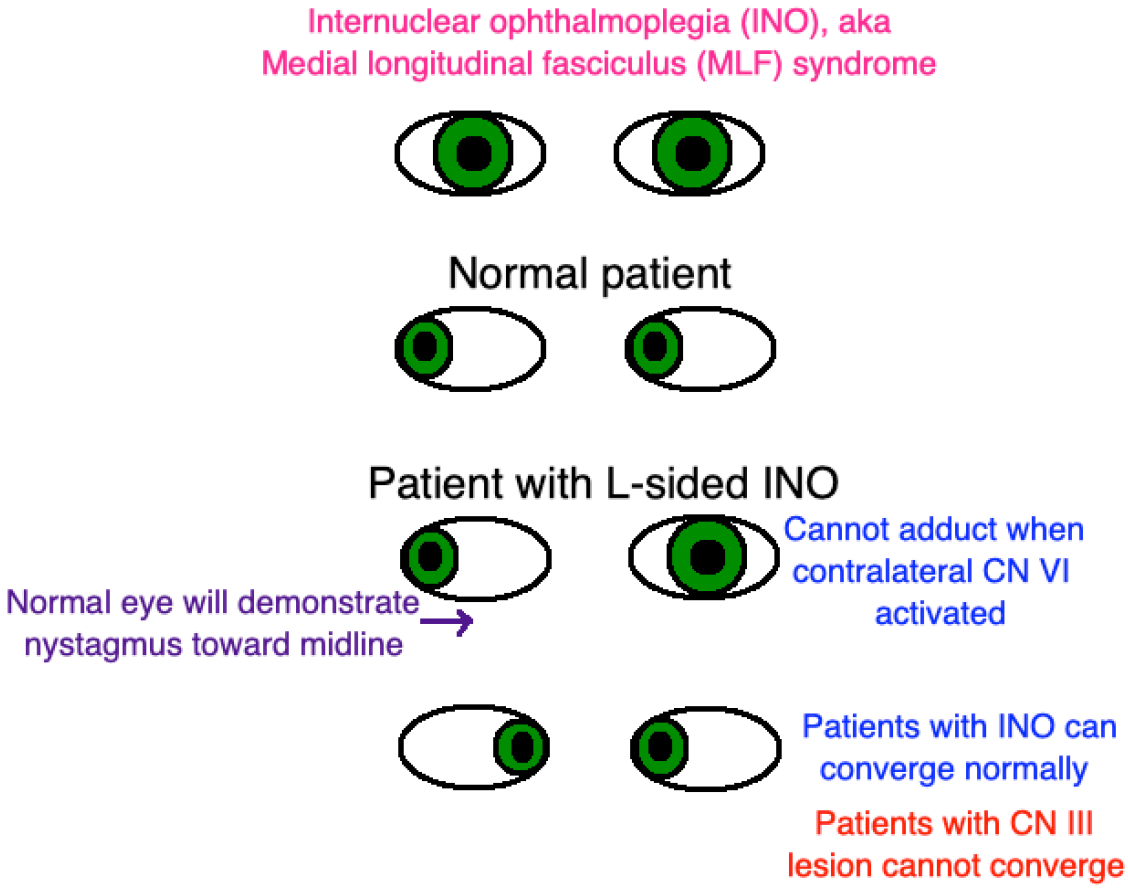
Parkinson disease / Parkinson-plus disorders	
- A Parkinson-plus disorder is a disease that presents similarly to Parkinson disease, but it's not.	
Parkinson disease	- Loss of dopamine-secreting neurons in the substantia nigra pars compacta of the midbrain, with deposition of alpha-synuclein on biopsy. - Presents with classic features of bradykinesia/akinesia, resting tremor, shuffling, short-steppage gait, micrographia, and cogwheel rigidity. - Alpha-synuclein gene mutation most common; many genes implicated. - Carbidopa-levodopa is combo frequently used for Tx. Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa.

	- Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on NBMEs, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer." - Be aware of D2 agonist names – i.e., ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine, which can all in theory be used as treatments. - Amantadine increases presynaptic release of dopamine. - Selegiline inhibits monoamine oxidase B, which is an enzyme that preferentially breaks down dopamine. USMLE wants you to know this can cause serotonin syndrome, either alone, or in combo with drugs like St John Wort or SSRIs.
RLS	- Restless leg syndrome. - Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used. - USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.
NPH	- As discussed earlier, normal pressure hydrocephalus presents as "wet, wobbly, wacky" +/- Parkinsonism. - The parkinsonism is the most overlooked detail by students, who will usually only know the wet, wobbly, wacky part. - What the USMLE will do is give you an older male who has WWW triad + they give you 2-3 more sentences describing what sounds like Parkinson disease, where you're like "What the hell? Is this Parkinson disease?" No. It's just NPH, which is a Parkinson-plus disorder, where it can look like Parkinson disease but it ain't. - Once again, due to impingement on the corona radiata, and the mechanism they want for urinary incontinence = "failure to inhibit the voiding reflex." - CT shows enlargement of the lateral ventricles.
Wilson disease	- Parkinsonism in a young patient is Wilson disease till proven otherwise. - Excessive copper accumulation in tissues, including the basal ganglia and liver, due to inability to excrete it into bile. - I discuss this stuff in more detail in the HY Gastro PDF.
Lewy body dementia	- As discussed before, this is a Parkinson-plus disorder. - The patient will have dementia + visual hallucinations + Parkinsonism.
Progressive supranuclear palsy	- Obscure condition that gets asked on 2CK. - You just need to know that 100% of questions will say "axial dystonia + Parkinsonism," where they'll just ask for diagnosis straight-up. - Axial dystonia is a type of muscle condition resulting in abnormal posture and movement of the spine and torso.
MPTP	- Aka synthetic heroin. - Just know it is a cause of Parkinsonism. - Shows up on an NBME. Students are like wtf?

Optic neuritis
- Inflammation of CN II (optic nerve). - Presents variably as change in visual acuity, color vision, central scotoma, etc. **- Basically synonymous with multiple sclerosis on USMLE.** I discuss MS more later. - Causes Marcus Gunn pupil (aka relative afferent pupillary defect). In this case, the intensity of light is under-sensed by the affected eye (afferent signal), so the resultant efferent CN III (oculomotor) parasympathetic output (back to the eyes for pupillary constriction) is attenuated; this means when you shine a light in the affected eye, both pupils constrict less in comparison to when the light is shone in the

unaffected eye; this **gives the mere impression** that the pupils dilate when light is shone in the affected eye. Still confusing? Take a look at the following illustration (assuming a lesion of the patient's right eye [left side of illustration]):



MLF syndrome
- Medial longitudinal fasciculus syndrome (MLF syndrome; aka internuclear ophthalmoplegia; INO). - Pathognomonic for multiple sclerosis. MS patients can also get optic neuritis (as mentioned above), but this is not pathognomonic for MS (i.e., it can be seen in other circumstances, such as with ethambutol or sildenafil use). - Mechanism for pathology: when you look to one side (let's say the right), CN VI on the right and CN III on the left are activated. This unison of activation is accomplished when the left MLF in the midbrain is activated. In the case of MLF syndrome, the right eye can abduct without a problem, but the left eye will fail to adduct, and the right eye will exhibit nystagmus back toward the midline. - What you need to remember is: in INO, the side that cannot adduct is the side that's fucked up. So if you look to the right and the left eye doesn't adduct, then the left MLF is where the lesion is. 

Autoimmune neuropathies	
Multiple sclerosis	- T-cell-mediated attack against oligodendrocytes and myelin-basic protein. - Idiopathic autoimmune disorder classically in white women 20s-30s who live far from the equator. - Relapsing-remitting type is most common, where the patient will have repeated episodes of neurologic disturbances over many years at varying intervals. - Highest yield findings are optic neuritis and INO, as discussed above. - Urge incontinence is exceedingly HY. Some students ask about overflow incontinence due to MS; I've never seen this on NBME. Urge incontinence is what shows up all over the NBME forms for MS. I discuss the incontinences in extensive detail in my HY Renal and Repro/Obgyn PDFs. - Clonus and (+) Babinski sign (UMN findings) also HY. - MRI is gold standard for diagnosis, showing scattered white matter lesions within the CNS (i.e., both brain and spinal cord).

	- CSF analysis shows IgG oligoclonal bands. - Tx is steroids for acute flares (IV methylprednisolone). - β-interferon is what USMLE wants between flares to decrease recurrences. - Baclofen (GABA-B receptor agonist) is used for spasticity.
Neuromyelitis optica	- Aka Devic disease. - Sounds exactly like MS but they will say CSF oligoclonal bands are negative, and then the answer is just "antibodies against aquaporin-4." - You say Wtf? Not my opinion. It's asked on an NBME form.
ALS	- Amyotrophic lateral sclerosis. - In ALS, the lateral corticospinal tracts (UMN) and anterior horns (LMN) degenerate. - USMLE will often give vignette of ALS and then the answer is simply "motor neurons." - There are two things you will see in all ALS Qs, making the Dx very easy: 1) The vignette will always have combination of UMN and LMN findings. Fasciculations, decreased reflexes, decreased tone, and muscle atrophy are classic for LMN. Babinski reflex, clonus, increased (brisk) reflexes, increased tone are classic for UMN. Serum CK can be elevated due to increased tone. 2) Must have **no** sensory findings. If any sensory abnormalities (i.e., paresthesias, numbness, etc.) are present in the vignette, ALS is the wrong answer. - Knowing this second point in particular will help on difficult NBME Qs, where you'll get a big paragraph + they say somewhere in the stem something about a sensory abnormality, and you can say, "Cool, not ALS." - For 2CK, next best step in Dx = "electromyography and nerve conduction studies."
GBS	- Guillain-Barre syndrome (aka acute inflammatory demyelinating polyneuropathy). - T-cell- and antibody-mediated destruction of Schwann cells and myelin of peripheral nerves. - In other words, MS = CNS/oligodendrocytes; GBS = PNS/Schwann cells. - Can be described as "segmental and inflammatory demyelination." - Presents as ascending paralysis and loss of deep tendon reflexes. - Can affect both upper and lower limbs. - Symptoms can start as tingling in the hands and/or feet. - USMLE can say "weakness of proximal and distal muscles in lower limbs and weakness of distal muscles in upper limbs." This implies that the weakness has already ascended in the legs but not yet in the arms. - Textbook cause is *Campylobacter jejuni*, but in almost all Qs they will not give a patient who has bloody diarrhea after a BBQ. In other words, you can't use this as a crutch. They will just give you the symptoms alone and you have to know it's GBS.
CIDP	- Chronic inflammatory demyelinating polyneuropathy. - Vignette will sound exactly like GBS, but symptoms will be present for months and progress slowly. - GBS, in contrast, develops quickly over days and peaks within weeks. - In other words, if you get a vignette where it sounds like GBS, but they say the symptoms have been present for 4 months, that's CIDP, not GBS. There's one NBME Q on it.
CMTD	- Charcot-Marie-Tooth disease. - Obscure peripheral nerve autoimmune disease that is the answer on USMLE if they give you high-arched feet (pes cavus), hammer toes, and foot drop.
Bell's palsy	- LMN lesion of CN VII (facial nerve). - Paralysis of ipsilateral facial muscles. - Usually idiopathic/autoimmune following viral infection, shingles (i.e., Ramsay-Hunt syndrome II), or Lyme disease. - If the Q gives you Bell's palsy and asks for next best step, the answer is do serology for Lyme disease. If this isn't listed (or the patient doesn't live in endemic area, such as Africa), choose "no further diagnostic studies indicated." Nerve conduction studies are wrong (this is on NBME). Bell's palsy is a clinical diagnosis.

	- IV steroids, followed by a 10-21-day taper of oral steroids is indicated to minimize the immune response. - Ipsilateral hyperacusis (due to paralysis of stapedius muscle) and/or loss of taste to the anterior 2/3 of the tongue due to their innervation by CN VII. - There is a Q on NBME where they say patient has loss of taste following trauma + neurologic exam shows no abnormalities → answer = olfactory nerve (CN I) palsy, not CN VII, due to cribriform plate fracture (loss of smell means loss of most taste). If the Q wanted CN VII lesion, they would give a concurrent Bell's palsy presumably, since loss of taste wouldn't happen in isolation with CN VII palsy. - 19 times out of 20, LMN causing Bell's palsy is what we see for facial nerve lesions. However, 1/20 are UMN, which results in loss of motor function of the contralateral middle and lower face, with the forehead spared.
PANDAS	- Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococci. - In rare cases, Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask what will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Sydenham chorea	- Autoimmune movement disorder seen as part of rheumatic fever. - Chorea = fast, purposeless, jerky movements. - Vignette will give child who has rheumatic fever and then ask about the cause of the movement problem, and the answer is just "autoimmune," or "autoimmune disorder of basal ganglia." - I discuss rheumatic heart disease in detail in the HY Cardio PDF.
Myoneural junction disorders	
Myasthenia gravis	- Autoantibodies against post-synaptic nicotinic acetylcholine receptors. - Classic vignette is female office worker in her 40s who has triad of 1) diplopia, 2) dysphagia/hoarseness, and 3) ptosis that worsens throughout the day. - Gets worse with recurrent stimulation of muscle (vignette might say the patient cannot perform upward gaze for 60 seconds). - Tensilon test is administration of edrophonium (short-acting acetylcholinesterase inhibitor) → patient experiences significant improvement in symptoms. - Tx = pyridostigmine (longer-acting acetylcholinesterase inhibitor). - MG can be a paraneoplastic syndrome of thymoma. - 10-15% of patients with MG have thymoma. - If MG is diagnosed, a chest x-ray should be performed to look for thymoma. If the x-ray is abnormal, a CT scan is ordered. - Removal of the thymoma in these patients can significantly improve/cure the MG.
Lambert-Eaton	- Autoantibodies against pre-synaptic voltage-gated calcium channels. - Presents as proximal muscle weakness that improves with activity (vignette might say the patient tries to get up from a chair a few times before finally being able to). - Tensilon test does not significantly improve Sx in comparison to MG. - Highest-yield point is that it is a paraneoplastic syndrome of small cell lung cancer.

Vertigo	
BPPV	- Benign paroxysmal positional vertigo. - Vertigo is the feeling of the room spinning. - Brief episodes of dizziness, usually 30-60 seconds, sometimes with vomiting. - Caused by a semicircular canal otolith (ear stone; aka otoconia). - Otoconia made of calcium carbonate are normally found within the utricle and saccule of the inner ear and play a role in detecting acceleration and motion. If one becomes dislodged and enters the semicircular canals, BPPV results. - What the USMLE likes to do is give an easy vignette of BPPV, and then the answer is just "otolith." Easy, but if student hasn't heard of it before they're like wtf?

	- Diagnosis is made via Dix-Hallpike maneuver. In this test, nystagmus is induced when the patient is rapidly moved from a seated position to lying down with the head tilted backward and turned to one side. If nystagmus occurs, Dx = BPPV. - Tx is Epley maneuver, which is a series of head movements aimed at moving the semicircular canal otolith back to the utricle, thereby curing the BPPV.
Vestibular neuritis	- Viral infection followed by vertigo. - Inflammation of the vestibular nerve (branch of CN VIII, vestibulocochlear nerve).
Labyrinthitis	- Viral infection followed by vertigo +/- tinnitus. - Inflammation of both the vestibular nerve and cochlear nerve. - This distinction of "VN is just vertigo whereas labyrinthitis is also tinnitus" is perpetuated on the internet, but I can tell you there is an NBME Q (that is repeated across forms) where they don't mention anything about tinnitus and the answer is labyrinthitis, where vestibular neuritis isn't listed. This is why I write +/- for tinnitus above. - The tympanic light reflex can be abnormal, which can reflect increased middle ear pressure (sometimes seen with viral infection). I've seen the NBME write in the stem "abnormal light reflex," which is a confusing point, since this could be misconstrued as referring to the eyes, not the ear. The vignette in which they mention this they also have multiple sclerosis as a wrong answer + mention the patient has a sibling with MS. So the Q is dumb/gotchya-style overall. The key sentence in this NBME Q is that the patient had nausea and reduced ability to eat foods past few days, which implies recent viral infection.
Meniere	- Waxing and waning, asymmetric tinnitus and vertigo that has a slowly progressive course over many years. - Has a familial component but no strict inheritance (i.e., the vignette can sometimes mention a relative with similar findings). - Due to defective endolymphatic drainage. - Can cause low-frequency hearing loss (i.e., patient says it is difficult to hear conversations at the dinner table with many people). This is in contrast to presbycusis, which is high-frequency hearing loss in elderly.
Drug-induced	- Aminoglycosides (i.e., gentamicin, amikacin), chemo agents like cisplatin, and loop diuretics (furosemide, ethacrynic acid) are known for ototoxicity. - Key point is that this ototoxicity need not be hearing loss and can be vertigo – i.e., the patient is receiving IV Abx for endocarditis treatment and feels like the room is spinning (ototoxicity due to gentamicin).

HY headache types	
Trigeminal neuralgia	- 11/10 lancinating, knife-like pain that occurs as episodes lasting usually <1 minute. - Classically brought on by minor stimuli, such as brushing one's hair/teeth, or a gust of wind. - Thought to be caused by trigeminal nerve irritation or impingement (i.e., at the exit points from the skull). - There is an NBME Q where they ask what part of the brain is fucked up (i.e., they don't list any nerves), and the answer is pons, since CN V originates from the pons. - Prophylaxis is carbamazepine. - There is no Tx for acute episodes since pain lasts such short duration. - Often confused with cluster headache if occurring in the V1 (ophthalmic branch of trigeminal nerve) distribution.
Cluster headache	- Male 20s-40s who wakes up from sleep with 11/10 lancinating, knife-like headache; often associated with ipsilateral lacrimation, rhinorrhea, and sometimes pupillary changes (I've seen NBME say ipsilateral miotic pupil). - Episodes last minutes to half hour (longer than trigeminal neuralgia), where the Q will say the guy wakes up from sleep + paces around his room until pain eventually goes away.

	- Prophylaxis is verapamil. - Tx for acute episodes is oxygen (bizarre, unless patient has O2 tank lying around).
Migraine	- Unilateral throbbing/pounding headache that lasts hours. - Can be associated with prodromal visual aura or sounds. - Tx is NSAIDs first, followed by triptans (e.g., sumatriptan). - Triptans are serotonin (5HT) receptor agonists. - Triptans are not prophylactic meds; they are only used as abortive therapy. - Prophylaxis is propranolol. - USMLE will give 30s female with hypertension who has migraine Hx, and the treatment is "beta blockade" for her HTN management. - Women who were previously prescribed sumatriptan for abortive therapy should take acetaminophen instead while pregnant. This is asked on an NBME. - Estrogen-containing contraceptives are contraindicated in patients who have migraine with aura.
Tension headache	- Bilateral, dull, band-like headache. - Treated with sleep and acetaminophen.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another NBME Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.

WKS and ataxia	
- Ataxia means lack of coordination of voluntary muscles presenting as unsteady gait and balance.	
Wernicke-Korsakoff syndrome	- Damage to primarily the mamillary bodies due to thiamine (B1) deficiency. - Wernicke encephalopathy = **A COW** = **A**taxia/**A**nterograde amnesia, **C**onfusion, **O**phthalmoplegia, **W**ernicke. - Korsakoff psychosis = retrograde amnesia; causes confabulations, which means making up stories about the past because of loss of memory regarding prior events. - Therefore, WKS = A COW + confabulations. - Application to USMLE is that alcoholics presenting to hospital with confusion, ataxia, or eye findings require thiamine. - One NBME Q asks what giving thiamine will help reduce risk of in alcoholic → answer = anterograde amnesia (apparently by reducing risk of confusion due to Wernicke), where retrograde amnesia isn't listed as answer.
General Ataxia	- USMLE wants you to be able to differentiate ataxia caused by cerebellar vs dorsal column lesions.

	- Dorsal column lesions cause a (+) Romberg test, meaning the patient falls over when standing with the eyes closed due to loss of proprioception. - Cerebellar lesions won't cause a (-) Romberg test almost always due to maintenance of proprioceptive capacity.
Ataxia-telangiectasia	- Autosomal recessive immunodeficiency condition that causes – you'd never guess it – ataxia and telangiectasias. - What USMLE will do is make the diagnosis obvious, and then the answer will just be "failure of double-stranded DNA break repair."

Cerebellar lesion signs	
Asterixis	- Aka "hepatic flap." Seen in liver disease. - Motor disturbance characterized by brief, sudden lapses of muscle tone, presenting as the patient demonstrating a "flapping" motion of the hands when he or she extends the arms and hands anteriorly.
Dysdiadokokinesia	- Impaired ability to perform rapid alternating movements, presenting as difficulty rapidly tapping a finger or quickly pronating and supinating the hand.
Intention tremor	- Kinetic tremor that worsens as the patient's hand approaches a target.
Truncal ataxia	- Tends to occur with more midline cerebellar lesions (i.e., of the vermis). - Can be seen, e.g., in pediatrics with medulloblastoma, where a kid has morning vomiting (often indicates brain tumor) and truncal ataxia.
Limb ataxia	- Tends to occur with more lateral cerebellar lesions. - An important rule is that cerebellar lesions cause ipsilateral deficits, unlike many brain lesions that cause contralateral limb findings.

Neuro conditions caused by vitamin deficiencies	
Vit B12	- As discussed earlier, causes subacute combined degeneration, with reversible dementia and/or systemic neurologic findings. - Hx of gastrectomy, pernicious anemia, veganism, or strict vegetarianism is buzzy. - If patient has B12 deficiency + an unrelated autoimmune disorder, e.g., vitiligo, Hashimoto, etc., the cause is probably pernicious anemia, since "autoimmune diseases go together," where having one autoimmune disease ↑ risk of others. - Patients need not have neurologic dysfunction in B12 deficiency. Students commonly have this misconception. - Causes ↑ MCV (megaloblastic anemia) and hypersegmented neutrophils. - If Q gives you older patient on tea and toast diet + high MCV + no other information, folate (B9) deficiency is more common than B12 deficiency (on FM).
Vit B1	- Can cause WKS, dry beri beri (neuropathy), and wet beri beri (dilated cardiomyopathy). - Two new 2CK NBME Qs assess thiamine deficiency causing neuropathy (dry beri beri) *post-gastrectomy*, which is bizarre, since we classically think of this as causing B12 deficiency. - A helpful detail that B1 deficiency is the answer post-gastrectomy instead of B12 is if they give *confusion*, which is characteristic of Wernicke encephalopathy, not subacute combined degeneration. - One of the Qs also mentions a (+) Romberg test for B1 deficiency as well. Everyone gets these Qs wrong because the notion of B1 deficiency as correct post-gastrectomy, as opposed to B12, is recondite and obscure.
Vit E	- Can cause peripheral neuropathy. - Weirdly an extremely HY vitamin deficiency on USMLE, despite it not seeming that common. Shows up in **cystic fibrosis questions.** - Patients with CF have exocrine pancreatic insufficiency leading to fat-soluble vitamin malabsorption. So CF patient + neuropathy = vitamin E deficiency. - Can in theory be seen with any cause of exocrine pancreatic insufficiency (i.e., chronic pancreatitis due to alcoholism, pancreatectomy, severe diabetes), but it's HY for CF Qs as I said.

	- Likewise, bowing of tibias / low Ca^{2+} in CF patients would be vitamin D deficiency.
Vit B6	- Neuropathy and/or seizures in patients taking isoniazid for tuberculosis. - Isoniazid causes B6 deficiency.

Tremor types for USMLE	
Essential	- Autosomal dominant intention tremor. - Can occur at any age, but usually middle age and older. - Patients self-medicate with alcohol, which ↓ tremor. - Propranolol (beta-blockade) is treatment.
Resting	- Parkinson disease (or Parkinson-plus disorders) in older patients. - Wilson disease till proven otherwise in younger patients.
Intention	- Cerebellar lesions or essential tremor.
Drug-induced	- Lithium and valproic acid can both cause tremor. - Lithium is classic for tremor, but an NBME Q gives valproic acid as correct answer (and lithium wrong answer) for patient who has tremor + abnormal liver function tests, since valproic acid is known for hepatotoxicity but lithium is not. - Albuterol (β2-agonist for asthma) shows up in an NBME Q as causing tremor. - Alcohol withdrawal (delirium tremens). Treat with benzo.

Meningitis	
- Presents as high fever + **stiff neck** and photophobia.	
Bacterial	- In neonates: Group B Strep (*Strep agalactiae*; gram-positive cocci) most common. Can also be *Listeria* (gram-positive rods) or *E. coli* (gram-negative rods). - Everyone else (not neonates): *S. pneumo* (gram-positive diplococci) and *N. meningitides* (gram-negative diplococci) most common. - CSF shows ↑ Protein, ↓ glucose, ↑ neutrophils (polymorphonuclear cells).
Fungal	- *Cryptococcus neoformans* in immunocompromised patients (i.e., HIV, or organ transplant recipients taking immunosuppressant meds) is most common. - CSF shows ↑ Protein, ↓ glucose, ↑ lymphocytes.
Viral	- Aka aseptic meningitis. - Commonly echovirus (if vaccinated); can also be mumps (if unvaccinated). - Can rarely even rarely be due to acute HIV infection. - CSF shows normal protein (or a tad elevated), normal glucose, ↑ lymphocytes.
Encephalitis	
- Presents as high fever + **confusion.**	
HSV 1/2	- Herpes encephalitis can occur following vertical transmission in neonates. Can also be previously young healthy adults. - **RBCs in the CSF** due to temporal lobe hemorrhage. - Non-contrast CT of the head is often negative, so don't be confused by this. However they can say EEG shows abnormal spikes over the temporal region. - Other CSF findings same as viral meningitis. - I've only ever seen one NBME Q where RBCs were absent in CSF in herpes encephalitis, but you could still get the answer because other CSF findings were viral + no other virus listed.
Step 2CK/3 only	
- Do antibiotics before lumbar puncture. - Four indications for doing CT before LP in suspected meningitis / encephalitis: 1) Focal neurologic signs. 2) Seizure. 3) Papilledema, or the optic fundi cannot be visualized. 4) Confusion that interferes with neurologic exam / reduced Glasgow score. - The reason for CT before LP in the above scenarios is because of the risk of intracranial mass lesion, where if LP is performed, which acutely ↓ intracranial pressure, herniation and death might occur.	

- 2CK NBME Q gives point #3 above, where answer is CT, not LP, and many students get it wrong.
- Empiric Tx for meningitis is ceftriaxone + vancomycin. Steroids can be added if confirmed *S. pneumo*.
- For Group B Strep in neonates only, choose ampicillin + gentamicin, or ampicillin + cefotaxime; the combo of vancomycin + ceftriaxone is wrong on the NBME for GBS.
- We usually avoid ceftriaxone in neonates and prefer cefotaxime as the 3rd-gen cephalosporin. The exact age of the transition is not so important. What you need to know is, if you get a Peds Q and both cefotaxime and ceftriaxone are listed, cefotaxime is what we generally prefer.
- Ceftriaxone, compared to cefotaxime, displaces bilirubin from albumin, thereby ↑ risk of neonatal hyperbilirubinemia. It can also cause calcium deposits in the lungs and kidney.
- HY that you know meningitis can cause sensorineural hearing loss. If they tell you, e.g., 6-year-old boy isn't paying attention in class + isn't doing well in school + has Hx of meningitis; next best step = air-conduction audiometry. Do not confuse with tympanometry, which evaluates mobility of the tympanic membrane and can be used in acute otitis media.

Other CNS infections	
Toxoplasmosis	- *Toxoplasma gondii*; protozoan (unicellular eukaryote). - Causes ring-enhancing lesion(s) on CT head. - Doesn't need to be in patient who has exposure to cats. - Can be acquired from pork. - Prophylaxis is trimethoprim-sulfamethoxazole (TMP-SMX) in HIV patients with CD4 count <100/μL. - Since patients with HIV are already commenced on TMP-SMX for *Pneumocystis jirovecii* pneumonia (PJP) prophylaxis once CD4 count is <200, the latter functions as two bird with one stone for eventual Toxo prophylaxis. - It is important for USMLE that you know the prophylaxis for PJP and toxo is the same, since if they give you ring-enhancing lesion on CT in HIV patient with CD4 count <100 who's on TMP-SMX, you know it's not Toxo. In this case, the diagnosis is primary CNS lymphoma. - The treatment for toxo is sulfadiazine + pyrimethamine. This distinction is asked on an NBME, where TMP-SMX is also listed and wrong.
AIDS	- Can cause AIDS complex dementia, as discussed earlier. This presents as "wet, wobbly, wacky" (similar to NPH) in an AIDS patient. - Can cause primary CNS lymphoma, especially when CD4 count is <100/μL.
Neurocysticercosis	- *Taenia solium*; cestode; pork tapeworm. - Causes lesions within the brain in patient who went abroad, e.g., to Mexico. - Treated with albendazole or praziquantel.
Mastoiditis	- Complication of untreated or severe otitis media. - Q will tell you a child has tenderness behind the ear + the pinna is displaced upward and outward. - Next best step is CT or MRI of the temporal bone to visualize a potential fluid collection, since if this is not drained it can lead to brain abscess. - NBME has CT of temporal bone as answer for mastoiditis; sounds wrong to do head CT in a kid, but it's what they want on the form. - X-ray is wrong answer, as it doesn't adequately visualize a potential fluid collection.
Brain abscess	- Abscess in the brain. Not rocket science. - Can be caused by septic embolus due to endocarditis. - Q will give you murmur + fever + neurologic findings → vegetation from heart valve has launched off to the brain, causing stroke-like presentation. - Tx = drainage of abscess + IV antibiotics.
Epidural abscess	- Usually presents as "point tenderness over a vertebra," which on USMLE means epidural abscess, compression fracture (osteoporosis), or metastases. - Patient may or may not have neurologic findings. Fever is not mandatory. - Three demographics I've seen on NBME exams for epidural abscess: 1) IV drug user who now has point tenderness over vertebra;

	2) Patient who's had some sort of cellulitis of the legs / lower body that has spread to the spine, where you've eliminated the other answers to get there; 3) Patient with active endocarditis (i.e., murmur + fever) who has neurologic findings in the legs (septic embolus also possible, but this tends to produce stroke-like findings rather than spinal pain + lower leg findings). - Tx = drainage of abscess + IV antibiotics.
Pott disease	- Infection / granulomatous destruction of the vertebral column by tuberculosis. - Just know that this is possible, the same way psoas abscess is also possible.
Rabies	- RNA virus that ascends peripheral nerves to CNS; near-100% fatality rate. - If patient comes into contact with rabid animal, give rabies IVIG + multiple courses of vaccine. - Qbank will focus on exact days/spacing to give vaccine. Waste of time/garbage.
Amoebic meningoencephalitis	- *Naegleria fowleri*; amoeba that lives in lakes, streams, hot springs that enters through the cribriform plate. - Causes rapid deterioration and death within days. - You'll see a news article on this every 6 months or so about some "brain-eating amoeba" that killed person who swam in lake.
Eosinophilic meningoencephalitis	- *Angiostrongylus* (rat lung worm). - Consumption of raw slugs/snails. - Caused paralysis and death in an Australian teenager who ate raw garden slug after being dared by friends.

Cavernous sinus thrombosis
- Severe headache, diplopia, proptosis, and/or ophthalmoplegia in patient with recent sinus infection. - Bacterial infection can ascend the valveless facial veins to the cavernous sinus, leading to clotting of blood. - Usually *S. aureus*. - Tx = IV antibiotics + heparin. - CN VI palsy can occur with pathologies of the cavernous sinus.

Primary CNS lymphoma
- HY and underrated diagnosis for USMLE. - Causes ring-enhancing lesion on head CT that can be mistaken for Toxo. As discussed above, if AIDS patient has CD4 count <100 and is already on TMP-SMX, the Dx is primary CNS lymphoma, not Toxo. - Can be seen in non-AIDS patients who have immunosuppression due to other causes (i.e., organ transplant recipients on immunosuppressant meds; patients undergoing radio- and/or chemotherapy). - Can be caused by autoimmune diseases. For example, if they give you a 1-liner where they say, "44-year-old woman with SLE has ring-enhancing lesion on CT" → answer = CNS lymphoma, not Toxo. - Many autoimmune diseases (e.g., SLE, dermatomyositis, Hashimoto) ↑ the risk of non-Hodgkin lymphoma.

Encephalopathy	
- General term that refers to altered brain function (usually presenting as confusion) due to various causes.	
Hepatic	- Seen in cirrhosis due to hyperammonemia. - Classically occurs on USMLE when there is an ↑ in blood in the GI tract, such as following acute variceal bleed. Blood is broken down in the GIT into proteins that liberate amino acids, which are converted by gut bacteria into ammonia that is absorbed. - Lactulose is important drug on USMLE for treatment of hepatic encephalopathy. It is a sugar that is broken down by gut bacteria into acidic products. This leads to the conversion of ammonia into ammonium. The latter is ionic and not readily absorbed. The USMLE wants "acidification; ammonium" as the answer for what happens when we give lactulose.

	- Neomycin is another drug for hepatic encephalopathy. It is an aminoglycoside antibiotic that kills ammonia-producing gut bacteria.
Uremic	- Confusion in the setting of renal failure due to elevated blood urea nitrogen.
Metabolic	- Confusion due to electrolyte imbalances, hypoglycemia, and thyroid storm.
Toxic	- Mental status changes due to toxins. Reye syndrome and lead poisoning are HY.
Inflammatory	- Herpes encephalitis (infectious). - Lupus cerebritis (autoimmune).
Anoxic	- Confusion caused by diminished oxygen delivery to the brain.

Neuropathic pain disorders	
Diabetic neuropathy	- Presents as pain, paresthesia, and/or numbness in diabetic. - USMLE likes TCAs (i.e., amitriptyline) or gabapentin as first-line for Tx. - Use nortriptyline in elderly if a TCA is given (fewer side-effects).
Herpetic neuralgia	- Pain, burning, tingling experienced prior to, during, or after herpes or varicella (shingles) episode. - Acyclovir is used to treat the virus; gabapentin used classically for the pain.
Post-trauma	- Patients who have injuries resulting in nerve trauma can sustain chronic neuropathic pain. Treat with gabapentin or TCAs. - Trauma can result in complex regional pain syndrome, which is neuropathic pain that occurs as sequela of injury that is disproportionate to the injury itself.
Thalamic pain syndrome	- Damage to the thalamus due to a stroke that results in body pain, usually contralateral to the side of the infarction, occurring many months post-stroke. - The answer on USMLE for severe limb pain many months following a stroke.

Cutaneous nerve endings + what they detect	
Pacinian corpuscles	- The answer for deep pressure and vibration.
Meissner corpuscles	- The answer for light touch and vibration.
Merkel disc	- The answer for detecting shapes and textures.
Free nerve endings	- The answer for pain and temperature.
Ruffini endings	- The answer for sustained pressure and skin stretching.

TNR + Prion disorders	
Huntington	- Autosomal dominant; CAG trinucleotide repeat (TNR) expansion on chromosome 4 resulting in neurodegeneration and choreoathetosis. - Chorea = fast, purposeless, jerky movements; athetosis = slow, writhing movements. - Anticipation = disorder occurs earlier and more severe with each successive generation; caused by expansion of the TNR repeat. - USMLE likes caudate nuclei as part of the brain that's fucked up.
Friedreich ataxia	- Autosomal recessive; GAA TNR disorder. - Causes ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, and early-onset type II diabetes.
CJD	- Creutzfeldt-Jacob disease. - Prion disease. A prion is a misfolded protein that causes the misfolding of other proteins. - CJD is abnormally folded protein within the CNS (PrP^{C} → PrP^{SC}), where the former is an α-helix that becomes a misfolded β-pleated sheet. Sounds absurdly pedantic, but it's known to be asked for whatever reason. - If the misfolded PrP^{SC} variant of the protein comes into contact with normal PrP^{C} variants, it causes them to misfold into additional PrP^{SC}, with the process functioning exponentially. - Presents as neurodegeneration over weeks to months + **myoclonus.**

	- The myoclonus is the highest yield detail, since you will sometimes see CJD as a distractor answer, where I'll say to the student, "The patient doesn't have myoclonus though."

Infective neuropathies	
Polio	- RNA virus that infects the anterior horns of the spinal cord and causes polio myelitis (inflammation of spinal cord). - Classically causes degeneration of motor neurons in one leg, leading to one leg much smaller than the other leg. - Highest yield points about Polio actually have zero to do with the infection and almost all to do with the vaccines. - Salk vaccine = Killed intramuscular. - Sabin vaccine = live-attenuated oral. - USMLE will ask what is a common feature of these vaccines that accounts for their efficacy? → answer = "neutralizing antibodies in the circulation." - Only the live-attenuated vaccine is capable of generating a CD8+ T-cell response and IgA secretion in the gut. - However, both the killed and oral are capable of generating IgG antibodies in the circulation. - If you're confused about the immuno, I talk about this stuff in my HY Immuno PDF.
PML	- Progressive multifocal leukoencephalopathy. - Caused by JC polyoma virus. - Presents as neurodegeneration over weeks to months in immunocompromised patient, i.e., AIDS patient with CD4 count <100, patients undergoing chemoradiotherapy, or those on immunosuppressant drugs. - USMLE wants you to know this condition is due to "reactivation of latent infection," which means the patient is infected at some point during life years ago, but the infection now manifests due to immunosuppression. "Acute infection in immunocompromised patient" is the wrong answer.
TSP	- Tropical spastic paraparesis. - Obscure neurologic condition caused by HTLV-1/2 (the same virus that causes mycosis fungoides and Sezary syndrome; if you are like wtf?, I discuss these in my HY Derm PDF). - Development of anti-neuronal antibodies that presents as neurodegeneration.
SSPE	- Subacute sclerosing panencephalitis. - Obscure condition characterized by cognitive decline and neuropathy due to reactivation of certain strains of measles in the CNS. - Will present as teenager experiencing cognitive decline + CSF analysis shows measles.

Toxin-induced neuropathies	
Botulism	- Causes flaccid paralysis. - Botulin toxin of *Clostridium botulinum* inhibits presynaptic SNARE protein, resulting in ↓ release of presynaptic acetylcholine, which is normally stimulatory at muscles. - Botulism on USMLE can present as constipation, hypotonia, and cranial nerve palsies. - Can also present as floppy baby syndrome. - USMLE can be weird about the answer, where I've seen them write on an NBME exam something along the lines of "prevents acetylcholine from binding to its receptor" as the MOA of the toxin, even though this isn't technically the direct effect. - Acquired as spores in honey in infants under 1; acquired as pre-formed toxin from canned goods in anyone older. - NBME exam wants you to know that administering the toxin does not change the strength of acetylcholine binding to its receptor. This is because the toxin isn't a competitive inhibitor + only ↓ endogenous ACh release; this has no impact on any ACh administered exogenously.
Tetanus	- Causes spastic paralysis.

	- Tetanus toxin of *Clostridium tetani* inhibits presynaptic SNARE protein, resulting in ↓ release of presynaptic neurotransmitters GABA and glycine, which are normally inhibitory. - Can present as opisthotonos (arched back) and trismus (lock-jaw). - Presents in two patients on USMLE: 1) Neonate born at home whose umbilical cord was cut with a kitchen knife + tied with twine; 2) Random dude who cuts himself in backyard. **Steps 2CK/3:** - DtaP given at 2, 4, 6, months, then again at 15-18 months, then again at 4-6 years. - School-age kids require Tdap booster at 11-12 years, followed by Td booster every 10 years thereafter. - Pregnant women should get Tdap at 27-36 weeks to protect neonate from pertussis. For cuts/wounds post-vaccine: - 0-5 years post-vaccine: no Tx is necessary. - 6-9 years post-vaccine: if clean wound: no Tx; if dirty wound: give Td booster. - 10+ years post-vaccine: if clean wound: give Td booster; if dirty wound: IVIG + vaccine. - In other words, only ever give IVIG if it's a dirty wound + has been 10+ years.
Tetrodotoxin	- Toxin acquired by pufferfish. - Inhibits sodium channels; causes paralysis and death. - USMLE might tell you pufferfish was eaten, and then the answer is just "inhibition of sodium channels."
Ciguatera	- Toxin acquired by various meaty fish. - Prevents sodium channel closure. - The answer on USMLE for temperature dysesthesia ("hot feels cold; cold feels hot").

Narcolepsy
- Chronic sleep disorder affecting the brain's ability to regulate sleep-wake cycles normally. - Characterized by excessive daytime sleepiness, sudden episodes of muscle weakness (cataplexy), sleep paralysis, and hallucinations. The latter tend to occur prior to sleep (hypnagogic) or upon waking up (hypnopompic). - Thought to be due to deficiency of a neurotransmitter called orexin (aka hypocretin), which normally promotes wakefulness. - Next best step in diagnosis is polysomnography (sleep study). - Treatment is modafinil (dopamine reuptake inhibitor + promotes release of orexin from hypothalamus).

Brain bleeds	
Epidural hematoma	- Rupture of middle meningeal artery (branch of external carotid); classically occurs following head trauma to the temple. **- Lucid interval** – i.e., patient loses consciousness briefly, followed by arousing and returning to normal (i.e., has a period of lucidity), then goes home thinking he/she is okay, then goes to sleep, then dies. - Bleed appears as lens (biconvex) shape; fast accumulating. - If low GCS, do "intubation + hyperventilation," *then* do craniotomy. If normal GCS, just do craniotomy. - Hyperventilation mechanism: decreased CO_2 causes decreased cerebral perfusion. This reduces intracranial pressure.

	
Subdural hematoma	- Rupture of bridging (superior cerebral) veins. **- No lucid interval** (i.e., patient did not lose consciousness). - Crescent-shaped; can be slowly accumulating and heterogenous in appearance due to combination of dried + fresh blood. - Increased risk in elderly/dementia and alcoholics; answer in acceleration-deceleration injuries (shaken baby syndrome; motor vehicle accidents). - NBME wants craniotomy as the answer for Tx. - Small subdurals can occasionally be observed, but I've never seen this as an answer on any of the NBME material. Qbank is not the same as NBME. - If the patient has decreased Glasgow score, do "intubation + hyperventilation" as answer before craniotomy. 
Subarachnoid hemorrhage	- Rupture of anterior communicating artery (AcoM) or posterior communicating artery (PcoM) saccular/berry aneurysm. - AcoM > PcoM in terms of location. - Patient often reports "worst headache of his/her life." - **Can present with stiff neck** (meningism) the same way meningitis does, due to irritation of the meninges by the hemorrhage. This often confuses students and is a very HY detail. - PcoM aneurysm is classically associated with ipsilateral blown pupil (mydriasis) due to CN III impingement.

	- HTN is most common risk factor in population and accounts for most cases. - Ehlers-Danlos and autosomal dominant polycystic kidney disease (ADPKD) are HY non-HTN-associated specific causes of saccular aneurysms. - A patient with HTN is still more likely to die from myocardial infarction than berry aneurysm (asked on NBME). But if patient has berry aneurysm, it is most likely from HTN. - Bleed appears like a “star fish” or “sand dollar.” Q can say there is blood visualized in the basal cisterns.  - Treatment is usually supportive. HTN must be controlled. Anticoagulants must be reversed (i.e., FFP for patients on warfarin). Nimodipine can prevent vasospasm and neurologic sequelae during convalescence.
Intracerebral hemorrhage	 - Rupture of Charcot-Bouchard microaneurysms within the lenticulostriate arteries in patients who have chronic HTN. - Can present on USMLE as a 1-2-line stem, without any image or patient info, where they just say, “X person has bleed + decorticate posturing; what’s the diagnosis?” → answer = intracerebral hemorrhage.

	- In other words, USMLE wants you to know intracerebral hemorrhage has high likelihood of causing brainstem compression, which can present as decorticate or decerebrate posturing. - Associated with Alzheimer (i.e., amyloid angiopathy, which presents in Alzheimer as recurrent hemorrhagic stroke). - Can be associated with brain cancer as well (e.g., glioblastoma multiforme).
Pituitary apoplexy	- Bleeding into the pituitary gland; can cause bitemporal hemianopsia. - USMLE wants you to know this can occur puerperium and also due to pituitary tumors. - NBME gives bitemporal hemianopsia in woman who just gave birth, where pituitary apoplexy is only answer listed that could cause this visual pattern.

Cushing reflex

- **Cushing reflex** is a triad seen with ↑ ICP, where the patient will have: 1) HTN, 2) bradycardia, and 3) respiratory disturbance (classically ↓ RR, but there's an NBME Q where they don't give RR as ↓).
- ↑ ICP causes compression of intracerebral vessels → chemoreceptors sense ↓ oxygen delivery + ↑ CO_2 → this causes sympathetic activation and vasoconstriction of peripheral arterioles → ↑ BP → ↑ stretch of carotid sinus baroreceptors → ↑ CN IX afferent firing to solitary nucleus of medulla → ↑ CN X efferent firing to nodal tissue → ↓ HR.
- The ↓ RR is due to ↑ ICP causing compression of brainstem.
- NBME will give you a big 15-line paragraph with patient who's been in an accident + has hypertension + ask you for the reason for his or her HTN → answer = "increased intracranial pressure."
- Don't confuse Cushing reflex with Cushing ulcer (completely unrelated), which is head trauma causing vagal outflow → increased acid production + peptic ulcers.

Glasgow coma scale (GCS)

Glasgow Coma Scale		
Response	Scale	Score
Eye Opening Response	Eyes open spontaneously	4 Points
	Eyes open to verbal command, speech, or shout	3 Points
	Eyes open to pain (not applied to face)	2 Points
	No eye opening	1 Point
Verbal Response	Oriented	5 Points
	Confused conversation, but able to answer questions	4 Points
	Inappropriate responses, words discernible	3 Points
	Incomprehensible sounds or speech	2 Points
	No verbal response	1 Point
Motor Response	Obeys commands for movement	6 Points
	Purposeful movement to painful stimulus	5 Points
	Withdraws from pain	4 Points
	Abnormal (spastic) flexion, decorticate posture	3 Points
	Extensor (rigid) response, decerebrate posture	2 Points
	No motor response	1 Point
Minor Brain Injury = 13-15 points; **Moderate Brain Injury** = 9-12 points; **Severe Brain Injury** = 3-8 points		

- It's to my observation across NBME exams (in particular 2CK Neuro CMS forms) that they will never ask you to formally calculate GCS score.

- What will occasionally surface in NBME Qs is “decorticate” or “decerebrate” posturing, where you have to know, “Oh that means GCS is low; patient has compromised brain function, possibly herniation / ↑ intracranial pressure (ICP).”
- And as I mentioned earlier regarding suspected meningitis diagnostic workup, if GCS is not 15, then head CT should be done before LP.

Post-concussion syndrome

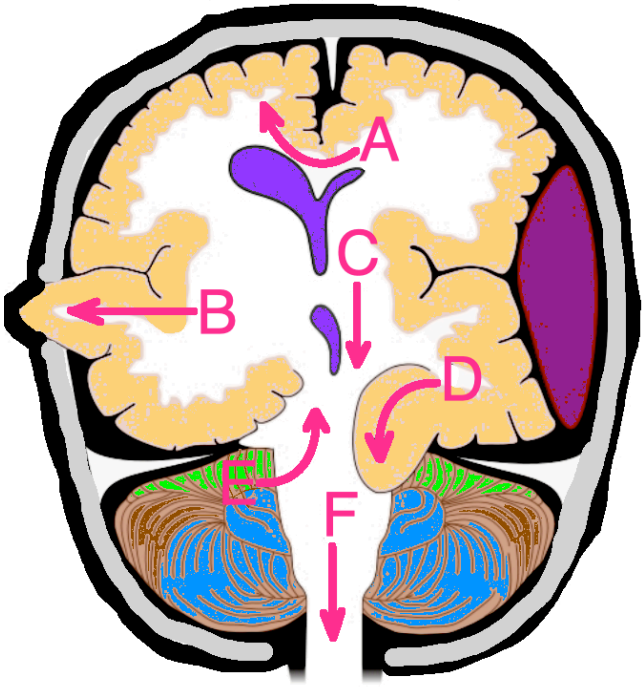
- A concussion is a type of traumatic brain injury caused by a blow to the head or by a sudden movement that results in the brain moving rapidly inside the skull. The rapid movement can cause brain cells to stretch and become damaged.
- Post-concussion syndrome (aka post-concussive disorder) comprises a range of symptoms that can persist for weeks, months, or even years after a concussion.
- Such symptoms include cognitive disturbance, memory loss, headache, and sensitivity to light or sound.

Base of skull fracture

- Fracture at base of the skull (you’d never have guessed it). Classic tetrad of signs is:
1) Battle sign, which is bruising over the mastoid process behind the ear.
2) Raccoon eyes, which refers to bilateral periorbital ecchymoses (bruising around both eyes).
3) Otorrhea (CSF leaking out of the ear).
4) Rhinorrhea (CSF leaking from the nose).
- Hemotympanum can also be seen, which is blood visualized behind the tympanic membrane.

Herniation types (2CK/3 Neuro only)

- Some students have asked if “2CK/3” is a typo above. It is not. Step 1 does not care that you are able to diagnose the specific types of brain herniations. But there is a difficult 2CK Neuro CMS Q that lists all different types of brain herniations, where you’re supposed to diagnose which one it is.



A	Cingulate (subfalcine)	- Doesn’t usually produce specific Sx but can compromise blood flow in the anterior cerebral artery, which is necessary for contralateral motor/sensory to the leg. - The cingulate gyrus shifts under the falx cerebri.

B	Transcalvarial	- Symptoms vary based on the herniated tissue, ranging from no symptoms to seizures or focal neurological deficits. - Brain tissue extrudes through a skull defect.
C	Central transtentorial	- Decreased level of consciousness and bilateral motor abnormalities. - The diencephalic structures move downward through the tentorial notch.
D	Uncal	- Ipsilateral mydriasis + contralateral motor deficits. - Involves the medial temporal lobe (uncus) shifting into the tentorial notch. - The ipsilateral mydriasis is due to CN III compression. The contralateral motor deficits are due to compression of the cerebral peduncle.
E	Upward transtentorial	- Decreased level of consciousness; pupillary changes. - The diencephalic structures move upward through the tentorial notch.
F	Tonsillar	- Life-threatening respiratory and cardiovascular compromise. - Cerebellar tonsils move downward into the foramen magnum, compressing the brainstem. Can cause decerebrate or decorticate posturing.

CNS malignancy	
- Can present with seizure and/or morning vomiting (a sign of ↑ ICP). - Can be described as ring-enhancing lesions (i.e., this descriptor is not limited to Toxo; it can also be tumors/metastases and abscesses).	
Peds brain tumors	
Pilocytic astrocytoma	- Most common brain tumor in kids. - Has solid (white) and cystic (black) components on imaging. - Rosenthal fibers on histo (eosinophilic corkscrew filaments). - Not aggressive.
Medulloblastoma	- Aggressive brain tumor of cerebellum in pediatrics. - Can cause truncal ataxia if located at vermis, and ipsilateral limb ataxia if lateral. - USMLE can give, e.g., morning vomiting in kid who has truncal ataxia, and the answer is medulloblastoma of cerebellar vermis.
Craniopharyngioma	- Most common pituitary tumor in children and teenagers. - Bitemporal hemianopsia (can occur with any pituitary lesion). - Embryo is that it originates from Rathke pouch, which will show up as “roof of the primitive oropharynx,” or “roof of the primitive oral cavity” as the answer. - Can calcify and contain cholesterol crystals; high rate of recurrence. - Don’t confuse that embryo with “endoderm of foramen cecum,” which refers to thyroglossal duct cyst.
Pinealoma	- Can cause vertical gaze palsy in kids (Parinaud syndrome).
Hamartomas	- The “tubers” in tuberous sclerosis are hamartomas, which are described as periventricular or cortical nodules on imaging.
Adult brain tumors	
Glioblastoma multiforme	- Most common adult brain tumor; aggressive. - Aka “butterfly glioma,” since it can cross through the corpus collosum. - Heterogenous in appearance due to areas of necrosis and hemorrhage. - USMLE wants you to know GM is a grade IV fibrillary astrocytoma.
Meningioma	- Benign tumor that grows from the meninges and appears to sit on top of the brain. - NBME Q floating around gives one that they describe as adjacent to falx cerebri. - *Can also occur within the spinal canal,* overlying the spinal cord (on NBME).
Schwannoma	- Arises from Schwann cells, which are responsible for myelinating peripheral nerves. - Vestibular schwannoma arises from CN VIII (vestibulocochlear nerve), often at the cerebellopontine junction, and can cause tinnitus and/or vertigo. - Can be idiopathic or as part of neurofibromatosis type II (NF2). - USMLE wants you to be able to identify its growth as causing sensorineural hearing loss (Rinne: AC > BC; Weber: lateralizes contralaterally).
Prolactinoma	- Most common pituitary tumor in adults. - Can cause amenorrhea in women and impotence in men.

	- Galactorrhea can occur in everyone. - Classically treated with bromocriptine (D2 receptor agonist), since dopamine inhibits prolactin secretion.
Optic glioma	- Seen in neurofibromatosis type I (NF1).
Oligodendroglioma	- Tumor of oligodendrocytes (the cells that normally myelinate CNS neurons). - Histo has fried-egg appearance and chicken-wire capillary pattern.
Ependymoma	- Tumor of ependymal cells, which normally produce CSF.
Tumor of foramen magnum	- Can cause hypoglossal nerve (CN XII) palsy, leading to ipsilateral tongue deviation. - Shows up on an NBME. - So you should know that CN XII lesions occur from medial medullary syndrome as well as tumors of the foramen magnum.
Metastases	- Can be described as lesions at the grey-white junction. - Choriocarcinoma and lung cancer love to metastasize to brain.

Tic disorders	
Tourette	- Neurologic disorder characterized by repetitive, involuntary movements and vocalizations (i.e., tics). - Motor tics range from mere eye blinking, head jerking, or shoulder shrugging to more complex movements such as hopping, twirling, or even mimicking others' actions. - Vocal (phonic) tics include throat clearing, sniffing, or humming. More complex vocal tics can involve repeating words or phrases, or shouting inappropriate words (coprolalia). - Haloperidol and clozapine are two agents used to Tx that show up on NBME.
Provisional tic disorder	- In pediatrics it is normal for the kid to sometimes have a singular tic that lasts up to a year (e.g., unusual tongue or lip gesture). - There is very low chance of it progressing to chronic tic disorder / Tourette. - USMLE wants follow-up in 3-6 months as the next best step (i.e., observe).
PANDAS	- As mentioned earlier, there is an obscure condition called Pediatric Autoimmune Neuropsychiatric disorder Associated with Streptococci, where Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask what will most likely diagnose etiology for the disorder → answer = “anti-streptolysin O titers.”
Drug-induced	- Stimulants such as methylphenidate, amphetamines, and caffeine can cause tics. - Tardive dyskinesia due to anti-psychotics is considered a type of tic disorder.

Seizure terminology	
FOAS	- Focal-onset aware seizure (FOAS). - Formerly known as simple seizures. - No loss of consciousness (LoC).
FOIAS	- Focal-onset impaired awareness seizure (FOIAS). - Formerly known as complex seizures. - LoC. - This includes staring into space blankly, as with absence seizures (discussed below).
Partial	- One part of the brain is affected. - Simple partial = no LoC + only affects one part of brain. - Complex partial = LoC + only affects one part of brain.
Generalized-onset	- Formerly known as generalized. - Dumb changes in nomenclature. But I don’t know what to tell you. - Involves both cerebral hemispheres.
Tonic-clonic	- Aka "grand mal" seizure; type of generalized seizure. It is characterized by two phases:

	- Tonic phase: lasts a few seconds; muscles stiffen and the patient falls to the ground and loses consciousness. - Clonic phase: rhythmic jerking movements of the limbs; usually lasts several minutes. - Following the seizure (i.e., during the postictal phase), the patient may be confused, drowsy, and have no recollection of the seizure. - Tongue biting/lacerations are a HY post-seizure finding. If the Q tells you explicitly that the tongue is normal in this setting, they are telling you it is not a seizure. - Tx for recurrent seizures are agents such as valproic acid, carbamazepine, and phenytoin. I discuss these later in this PDF in the pharm section. - There is a difficult NBME Q where they tell you a patient has twitching of one arm prior to falling to the ground and having a tonic-clonic seizure. The answer on the NBME form is "complex partial," not generalized tonic-clonic. This type of seizure is called FOIAS with secondary generalization, or complex partial with secondary generalization. - In other words, if a patient has focal neurologic signs preceding a tonic-clonic seizure, this indicates an origin in one location prior to spreading to other areas.
Status epilepticus	- The updated definition is a seizure lasting >5 minutes, or 2 seizures within 5 minutes. - Definition used to be a seizure lasting >30 minutes, or 2 seizures within 30 minutes. - First-line Tx is a benzo (e.g., IV lorazepam). - If benzo doesn't work, phenytoin (or fosphenytoin) is next, followed by barbiturates. - In other words, for USMLE: benzo → phenytoin → barbiturate (e.g., phenobarbital).
Myoclonic	- A type of generalized seizure that causes muscle jerks lasting less than a second. - A succession of jerks can be seen over a short time period. - The patient will not have loss of consciousness. - Can present similarly to simple partial, except the EEG will show generalized activity.
Febrile seizure	- Fever can precipitate idiopathic seizure in 2-4% of children ages 6 months - 5 years. - About a two-fold risk progression to epilepsy compared to general population. - Tx is with benzodiazepine. - Febrile seizures lasting longer than 10 minutes, seizures that are recurrent within 24 hours, or focal neurologic signs more significantly ↑ risk of progression to epilepsy.
Absence seizure	- Vignette will be a kid staring off into space in class for 30 seconds spacing out, sometimes with rapid blinking. - This is considered loss of consciousness. - EEG shows symmetric 3-Hz spike-and-wave discharges. - Tx is ethosuximide (thalamic calcium channel blocker).

Epilepsy disorders	
Temporal lobe epilepsy	- Most common epilepsy disorder. Originates at the medial temporal lobe. - Seizures are often preceded by visual auras, or warning signs. These can manifest as gustatory/olfactory sensations or déjà vu. - FOIAS are most common type, with or without secondary generalization. - For USMLE, pick "medial temporal lobe," or just "temporal lobe," if they ask for the origin of a seizure in the absence of any preceding neurologic findings.
West syndrome	- Aka infantile spasms; X-linked recessive. - Epilepsy syndrome in infants characterized by – you'd never guess it – spasms. - Causes an abnormal EEG pattern called hypsarrhythmia, which is chaotic pattern. - Leads to intellectual disability. - Treatment is with ACTH (obscure, but apparently ↑ endogenous cortisol, which can mitigate the progression).
Lennox-Gastaut	- Severe childhood-onset epilepsy with near-daily seizures and cognitive decline. - Hyperoralism is HY sign of cognitive regression (babies put things in their mouths). - Poor prognosis.
Juvenile myoclonic epilepsy	- Characterized by myoclonic jerks (usually hypnagogic and/or hypnopompic) that progress to tonic-clonic seizures after several months.

	- Age of onset is usually 10-16, hence juvenile, LOL! - Tx is valproic acid.

Conditions misconstrued as seizures	
Adams-Stokes attack	- Idiopathic arrhythmia disorder in kids that causes transient hypoxia to the brainstem, resulting in seizure-like episodes. - EEG does not show any seizure activity. - USMLE Q will give you kid with random arrhythmia + describe simultaneous twitching that sounds like seizure → answer = Adams-Stokes attack.
Breath-holding spell	- They'll say 3-year-old was having a tantrum followed by falling on the floor + appearing blue. - Child will involuntarily stop breathing following a trigger – i.e., being upset, frightened, or experiencing pain. - The child often cries or becomes upset, exhales forcefully, stops breathing, then develops cyanosis, which can sometimes be followed by brief loss of consciousness and jerking (can be mistaken for seizure). - Shows up on Peds forms so you need to know it exists.
Syncope	- Just be aware that syncope in elderly (e.g., from aortic stenosis or atrial fibrillation) can sometimes cause twitching of muscles that accompanies fainting, where it resembles a seizure. - The muscular activity is due to hypoxia to the brain, similar to Adams-Stokes attack. - Shows up on an NBME form as elderly patient with cardiovascular Hx who has seizure-like episode. - Vignette might say there is no evidence of tongue-biting (meaning not a seizure).

HY Peds neuro conditions	
Spinal muscular atrophy	- Aka Werdnig-Hoffman syndrome. - Genetic disorder affecting anterior horns. - Presents with profound hypotonia, absent reflexes, and **tongue fasciculations.**
Cerebral palsy	- Motor dysfunction and intellectual disability of varying severity, posited to be due to intrapartum conditions such as hypoxia. - Will show up on USMLE as spastic paraparesis in a child who has "scissoring" of the legs.
Chiari 1	- Downward displacement of the cerebellar tonsils into the foramen magnum. - Usually teenager or young adult. - Presents with headaches and neck pain, but can also be asymptomatic.
Chiari 2	- Aka Arnold-Chiari malformation. - Herniation of both the cerebellum and brainstem through the foramen magnum and is commonly associated with meningomyelocele. - Presents as a child with dysphagia and motor dysfunction.
Dandy-walker	- Absent cerebellar vermis + cystic dilation of 4th ventricle.
Fetal alcohol syndrome	- Most common cause of mental retardation. - Heart/lung defects (defective neural crest migration). - Thin vermillion border (thin upper lip). - Widely spaced eyes (hypertelorism), midface hypoplasia, short nose. - **Smooth, flat, and/or elongated philtrum.** - The USMLE can give you a vignette where the findings sound exactly like Down syndrome (i.e., kid born to 40-yr-old woman with slanted palpebral fissures + single palmar crease, etc.), but if they say *anything about the philtrum*, the answer is fetal alcohol syndrome, not Down syndrome. Shows up tricky on Psych forms like this.

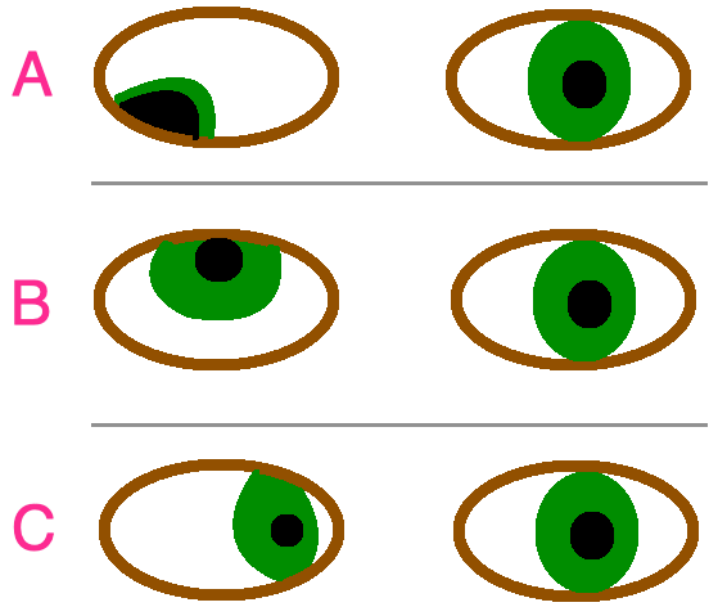
Phenylketonuria	- Autosomal recessive; deficiency of phenylalanine hydroxylase (or rarely THB, which is cofactor for the enzyme), causing inability to convert phenylalanine into tyrosine. - Screened for on heel-prick test at birth, since failure to detect PKU will result in mental retardation. - Phenylalanine-deficient diet must be given. - Presents as child with "mousy" or "musty" body odor + appears lighter skin than siblings (can cause partial albinism, since tyrosine is normally converted into melanin, and tyrosine can't be synthesized).
Maple syrup urine disease	- Autosomal recessive; deficiency of branched-chain ketoacid dehydrogenase. - Inability to breakdown leucine, isoleucine, and valine. - Causes a maple syrup-scent in the diapers from the urine. - Mental retardation and neurologic abnormalities if untreated.
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent exacerbation.
Waardenburg syndrome	- Congenital sensorineural hearing loss + pigmentary changes of hair and skin. - Shows up on NBME as young child with "white forelock" of hair. - Answer = failure of neural crest migration.
Double cortex syndrome	- Obscure condition asked on USMLE students are hysterical about. - Genetic condition where neurons are unable to migrate outwardly to reach their destination in the cortex, resulting in two grey matter layers (cortex) instead of one. - Mechanism USMLE wants is "failure of radial migration."
Rett syndrome	- Neurologic degeneration that occurs exclusively in female infants. - "Hand-wringing" or "hand-flapping" is buzzy finding. - Mental regression can present as putting objects in one's mouth (something babies normally do).
Cri-du-Chat	- Deletion on short arm of chromosome 5. - Results in distinctive cat-like cry during infancy, which is the source of the syndrome's name (French for "cry of the cat"). - Mental retardation, distinctive facial features, and microcephaly.
Friedreich ataxia	- As mentioned earlier, autosomal recessive GAA TNR disorder. - Causes ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, and early-onset type II diabetes.
Angelman	- "Happy puppet" – i.e., child presents with laughter. - Paternal imprinting → gene coming from dad is normally silenced; mom's gene is supposed to be expressed but is deleted or mutated. **- "Mom is not an angel"** because mom's gene is deleted. - Even if dad's gene is healthy, since it is normally silenced, it will not be expressed. - Can also be caused by paternal uniparental disomy. Since the gene is normally paternally imprinted, if both alleles are inherited from dad, both will be silenced; disease occurs because there's no maternal allele available to be expressed.
Prader-Willi	- Mental retardation + hyperphagia. - Maternal imprinting → gene coming from mom is normally silenced; dad's gene is supposed to be expressed but is deleted or mutated (**"Willi hates his dad"** because dad's gene is deleted). - Even if mom's gene is healthy, since it is normally silenced, it will not be expressed. - Can also be caused by maternal uniparental disomy. Since the gene is normally maternally imprinted, if both alleles are inherited from mom, both will be silenced; disease occurs because there's no paternal allele available to be expressed.

Phakomatoses (neurocutaneous disorders)	
Neurofibromatosis type I (NF1)	- *NF1*; AD; chromosome 17. - Neurofibromas, café au lait spots (hyperpigmented macules), axillary/groin freckling, Lisch nodules (iris hamartomas), pheochromocytoma, **optic nerve glioma,** oligodendroglioma, ependymoma.
Neurofibromatosis type II (NF2)	- *NF2*; AD; chromosome 22. - In kids, sometimes causes bilateral cataracts. - Bilateral acoustic schwannomas + meningioma in adults.
Tuberous sclerosis (TSC)	- *TSC1/2* genes; AD. - Periventricular nodules (tubers) on MRI of head, seizures, adenoma sebaceum (angiofibromas), subungual fibromas, renal angiomyolipoma, cardiac rhabdomyoma, hypopigmented macules ("ashleaf spots"), hyperpigmented velvety lesions ("shagreen patches").
Von Hippel-Lindau	- *VHL*; AD; chromosome 3. - Cerebellar/retinal hemangioblastomas, bilateral renal cell carcinoma, pancreatic cysts. - Mutation causes constitutive activation of hypoxia-inducible factor.
Sturge-Weber	- Not inherited; somatic mosaicism of *GNAQ* gene. - Port wine stain birth mark (nevus flammeus; may also present as violaceous papules in temporal distribution), leptomeningeal angioma (presenting as seizure); glaucoma.

Lysosomal storage diseases			
- All are associated with cognitive/neurologic deficits. - AR except for Fabry and Hunter, which are XR.			
Disease	**Enzyme deficiency**	**Buildup product(s)**	**HY points**
I-cell	N-acetylglucosamine-1-phosphotransferase	Many lysosomal enzymes	- Cannot synthesize mannose-6-phosphate at the Golgi; M6P needed to target lysosomal enzymes to the lysosomes; instead, lysosomal enzymes are secreted into cytosol + serum. - Course facial features, joint contractures, hepatomegaly, recurrent ear infections.
Gaucher	Glucocerebrosidase	Glucocerebroside	- Avascular necrosis of the hip. - "Crumpled tissue paper" (i.e., lipid-laded) macrophages. - The answer if they give you lysosomal storage disease + a bone problem.
Tay-Sachs	Hexosaminidase A	GM_2 ganglioside	- Cherry red spot on macula (blindness). **- No hepatosplenomegaly.** - Neurodegeneration.
Niemann-Pick	Sphingomyelinase	Sphingomyelin	- Cherry red spot on macula (blindness). **- Yes, hepatosplenomegaly.** - Niemann-Pick is a longer name than Tay-Sachs, and hepatosplenomegaly is a long word, so Niemann-Pick is the one with HSM. - Neurodegeneration.
Metachromatic leukodystrophy	Arylsulfatase A	Cerebroside sulfate	- Progressive neurologic decline.
Krabbe	Galactocerebrosidase	Galactocerebroside, psychosine	- Aka Globoid cell leukodystrophy. - Globoid cells (giant, multinucleated cells). - Neurodegeneration.
Fabry **(XR)**	α-galactosidase A	Ceramide trihexoside	- Angiokeratomas (small, red/purple dots on the skin due to capillary dilation).

			- Heart and/or renal problems.
Hunter **(XR)**	Iduronate sulfatase	Glycosaminoglycans (GAGs) – i.e., Dermatan + heparan sulfates	- Gargoyle-like facies, stridor, aggressive behavior. **- No clouded corneas.** - "Hunters can see the X" → Therefore XR and no clouded corneas.
Hurler	α-L-iduronidase		- Gargoyle-like facies, stridor, aggressive behavior. **- Yes, clouded corneas.** - More severe form of Hunter.

Extraocular nerve palsies



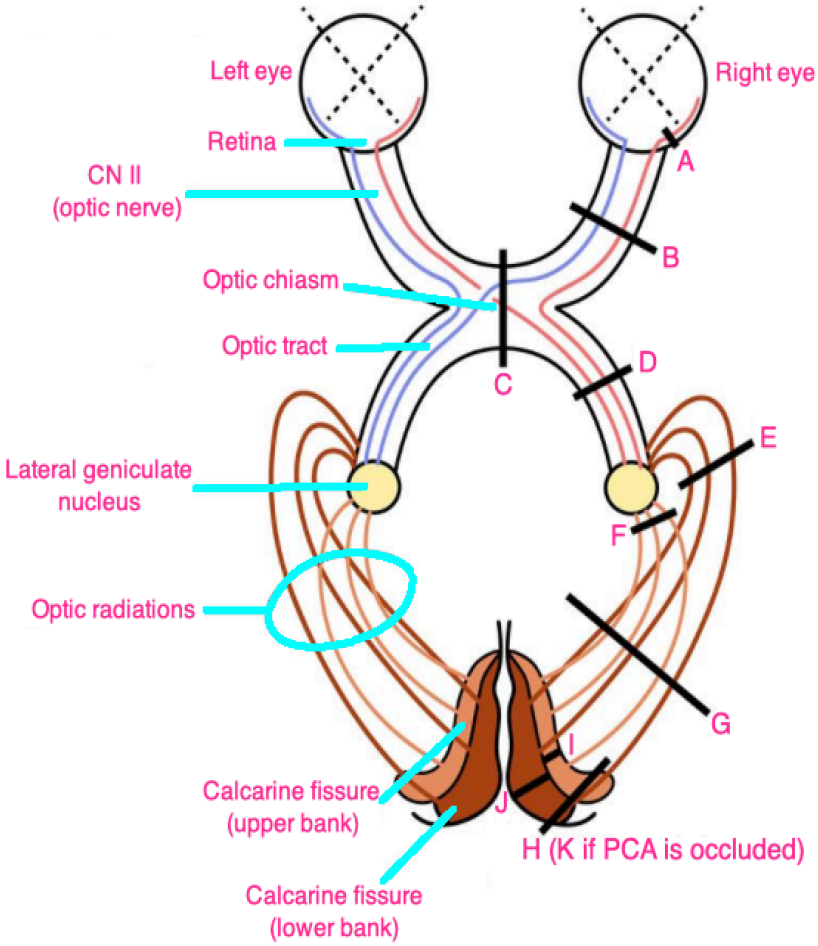
- A = CN III (oculomotor) lesion → presents as down and out. Can be idiopathic or in Weber syndrome.
- B = CN IV (trochlear) lesion → presents as upward gaze + patient will tilt head toward unaffected side.
- C = CN VI (abducens) lesion → medially directed eye. Commonly seen with cavernous sinus thrombosis.

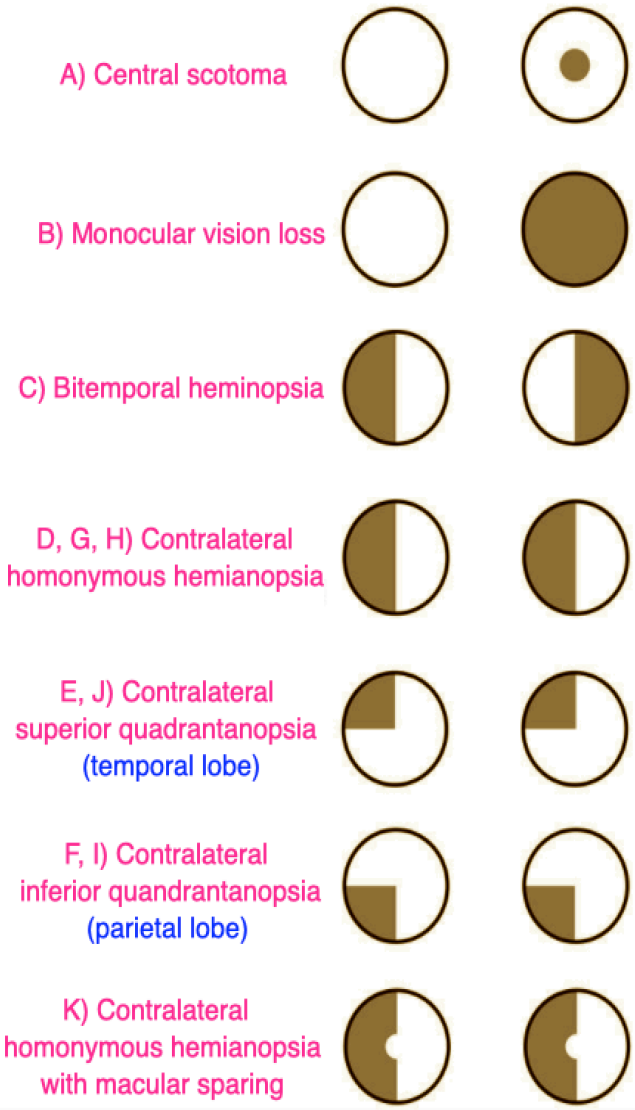
Cranial nerve exit points

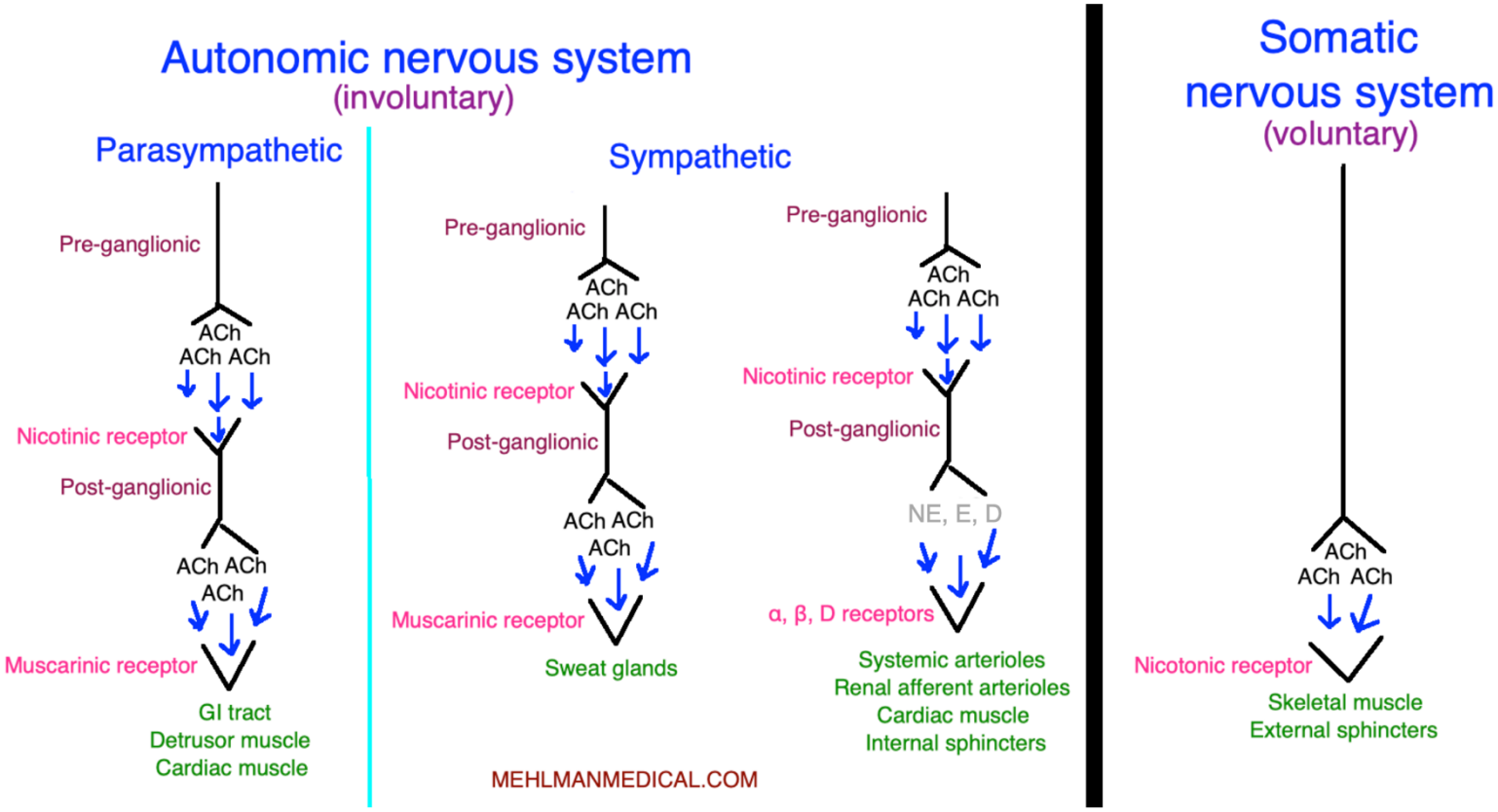
Cranial nerve	Exit point	Extra points
Olfactory (CN I)	Cribriform plate	- Only CN that doesn't relay through thalamus. - Injury causes impaired taste, since smell is required for taste. The vignette need not mention head trauma. If they don't give concurrent Bells palsy, you know the answer is olfactory, not facial nerve, since injury of the latter would also cause Bells palsy, not just isolated loss of taste.
Optic (CN II)	Optic canal	**- Afferent component of light reflex.**
Oculomotor (CN III)	Superior orbital fissure	**- Efferent component of light reflex (constricts pupil via parasympathetic innervation).**
Trochlear (CN IV)	Superior orbital fissure	- Innervates superior oblique (eye looks inferomedially).
Trigeminal ophthalmic branch (V1)	**S**uperior orbital fissure	- Afferent component of corneal reflex.
Trigeminal maxillary branch (V2)	Foramen **R**otundum	- Trigeminal mnemonic: "**S**tanding **R**oom **O**nly."
Trigeminal mandibular branch (V3)	Foramen **O**vale	- Muscles of mastication.

Abducens (VI)	Superior orbital fissure	- Abducts eye (patient gazes laterally).
Facial (VII)	Auditory canal	- Efferent component of corneal reflex. - LMN results in ipsilateral Bells palsy. - UMN results in contralateral mid + lower facial paralysis (forehead spared).
Vestibulocochlear (VIII)	Auditory canal	- Can give rise to acoustic schwannoma.
Glossopharyngeal (IX)	Jugular foramen	- Afferent component of gag reflex + carotid sinus baroreceptors.
Vagus (X)	Jugular foramen	- Efferent component of gag reflex + parasympathetic to slow HR; lesion causes **contralateral** uvula deviation.
Accessory (XI)	Jugular foramen	- Innervates sternocleidomastoid + trapezius.
Hypoglossal (XII)	Hypoglossal canal	- Lesion causes **ipsilateral** tongue deviation. - Tumor of the foramen magnum can cause CN XII palsy. - Medial medullary syndrome can cause CN XII palsy.

Rinne + Weber tests	
Rinne test	- A vibrating tuning fork is placed against the mastoid bone (bone conduction; BC). Once the patient can no longer hear it, it is removed from the bone and placed in front of the ear canal (air conduction; AC). - For both normal hearing and sensorineural hearing loss, AC > BC. - For conductive hearing loss, BC > AC.
Weber test	- A vibrating tuning fork is placed on the center of the forehead or top of the head, and the patient is asked if they hear the sound equally in both ears or if it's louder in one ear. - If it is louder in one ear, this is called “lateralization” to that ear.
Normal hearing	- Rinne shows air > bone conduction; Weber does not lateralize.
Conductive on left	- Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive on right	- Rinne shows bone > air conduction; Weber lateralizes to right.
Sensorineural on left	- Rinne shows air > bone conduction; Weber lateralizes to right.
Sensorineural on right	- Rinne shows air > bone conduction; Weber lateralizes to left.


Left eye
Right eye
Retina
CN II
(optic nerve)
Optic chiasm
Optic tract
Lateral geniculate
nucleus
Optic radiations
Calcarine fissure
(upper bank)
Calcarine fissure
(lower bank)
A
B
C
D
E
F
G
H (K if PCA is occluded)
I
J


A) Central scotoma
B) Monocular vision loss
C) Bitemporal heminopsia
D, G, H) Contralateral
homonymous hemianopsia
E, J) Contralateral
superior quadrantanopsia
(temporal lobe)
F, I) Contralateral
inferior quandrantanopsia
(parietal lobe)
K) Contralateral
homonymous hemianopsia
with macular sparing



Cholinergic receptors
- Called cholinergic because they bind acetylcholine (ACh). - Muscarinic and nicotinic types. - Muscarinic ACh receptors are G-protein-coupled receptors. If you want greater detail about which types of muscarinic receptors relate to which G-proteins, I talk about that stuff in my HY Biochemistry PDF. - Nicotinic ACh receptors are ligand-gated ion channels. When ACh binds, Na^+ and Ca^{2+} move into the cell. - On USMLE, "cholinergic" and "muscarinic" are used interchangeably virtually 19 out of 20 times. That is, when you hear "anti-cholinergic side-effects," or "pro-cholinergic effects," the meaning that should come to mind is primarily the effects of muscarinic receptors. - If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms –> **D**iarrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating. - If an agent is anti-cholinergic, it will cause the opposite of DUMBBELSS, or "anti-DUMBBELSS" –> Constipation, Urinary retention, Mydriasis, Tachycardia, Bronchodilation (in reality, β2 agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis. - Neuromuscular excitation is the one that is nicotinic. - The reason knowing this stuff is important is because TCAs (e.g., amitriptyline), antipsychotics, and 1st-generation H1 blockers (i.e., diphenhydramine, chlorpheniramine) have strong anti-cholinergic side-effects, which means they can cause "anti-DUMBBELSS" in any patient. In addition, anti-cholinergic effects promote delirium and confusion, especially in the elderly. - For example, old dude + high creatinine + on amitriptyline = post-renal azotemia due to a combo of BPH (which all old dudes have) and the TCA, which is anti-cholinergic. You want to take him off any anti-cholinergic meds he's on. Likewise, if you get a psych Q where a patient has confusion/delirium and one of his or her drugs is diphenhydramine, the answer might be, "discontinue anti-cholinergic medications."

- Cholinergic agents either: 1) agonize cholinergic receptors directly, or 2) increase the potentiation of ACh binding to its receptors by inhibiting acetylcholinesterase (the enzyme that breaks down ACh), thereby increasing Ach within the synaptic cleft.
- Almost all anti-cholinergic agents block cholinergic receptors directly.

Pro-cholinergic meds	
Muscarinic receptor agonists	
Bethanechol	- Stimulates detrusor muscle. - Used for overflow incontinence in diabetic patients with neurogenic (i.e., hypo-contractile) bladder.
Carbachol	- Constricts pupil and lowers intraocular pressure. - Used for glaucoma. - See my HY Ophthalmology PDF for more detail on that stuff.
Pilocarpine	- Constricts pupil and lowers intraocular pressure. - Powerful agent used for closed-angle glaucoma.
Methacholine	- Causes bronchoconstriction. - Used to help diagnose asthma. If administered to patient (not during acute attacks), spirometry will worsen and asthma symptoms may be replicated.
Acetylcholinesterase inhibitors	
Donepezil	- Increases acetylcholine available in synaptic cleft (i.e., pro-cholinergic). - Used 1st-line for Alzheimer disease. - Galantamine and rivastigmine have same MOA and use-case, but LY.
Edrophonium	- Aka Tensilon (the brand name). - Short-acting acetylcholinesterase inhibitor. - Administration to patients with myasthenia gravis causes marked improvement in symptoms and muscle function. - Administration to patients with Lambert-Eaton syndrome does not result in as marked improvement of symptoms.
Pyridostigmine	- Treatment for myasthenia gravis.
Physostigmine	- Antidote for Jimson weed poisoning (asked on NBME). - Jimson weed contains atropine-like compounds that are anti-cholinergic.
Neostigmine	- Used post-surgically to reverse the effects of neuromuscular blockade. - Toward the end of surgery, administration helps increase availability of ACh so that it can bind to its receptors and displace the nicotinic receptor antagonists used for neuromuscular paralysis.
Organophosphates	- Vignette will either be backpacker in his/her 20s picking fruit on a farm, or someone who drank fluid in a suicide attempt. - The phosphate group binds to acetylcholinesterase, inhibiting it. - NBME Q asks how toxicity could best be prevented → answer = "wearing gloves," where "wearing mask" is wrong answer. So acquisition occurs mostly through skin. - USMLE wants atropine first, followed by pralidoxime as Tx (discussed below).

Anti-cholinergic meds	
Muscarinic receptor antagonists	
Atropine	- Classic textbook drug that blocks muscarinic receptors. - Can be used to increase heart rate during surgery. - Given as 1st-line Tx for organophosphate poisoning before pralidoxime. - Pralidoxime is a special anti-cholinergic that works by regenerating functional acetylcholinesterase by kicking out the phosphate group that inhibited it.
Benztropine	- Used for acute dystonia due to anti-psychotics.

	- In other words, if patient starts anti-psychotic and then within hours to days gets stiff neck (torticollis), muscle rigidity (without fever), and/or abnormal eye movements (oculogyric crisis), this is 1st-line Tx. - Muscle rigidity with fever, in contrast, suggest neuroleptic malignant syndrome.
Diphenhydramine	- 1st generation H1-blocker that has strong anti-cholinergic side-effects to the point that it is often just characterized straight-up as a primary anti-cholinergic med. - Can be used same as benztropine for acute dystonia (USMLE won't list both). - Can be used for motion sickness (anti-cholinergic effects treat motion sickness). If USMLE tells you it is used in this setting + they ask you antagonism of which receptor enables its anti-motion sickness effect, the answer is M3. The wrong answer is H2, since this refers to stomach acid secretion. Diphenhydramine is an H1 blocker, not H2. - Most Qs on USMLE regarding diphenhydramine revolve around you knowing it should be discontinued from a patient's regimen, rather than added.
Prochlorperazine	- Anti-psychotic used as an anti-nausea med. - Even though it is an D2-receptor antagonist, as I mentioned before, anti-psychotics have anti-cholinergic side-effects. In this case, the anti-muscarinic side-effects are actually a "good" thing since these treat nausea and motion sickness.
Ipratropium	- One of the 1st-line agents used for COPD (asked on 2CK FM forms). - COPD patients can be started on either a muscarinic receptor antagonist, such as ipratropium, or a beta-2 agonist, such as olodaterol. - New literature gets pedantic about the duration (i.e., long- vs short-acting) anti-muscarinics and beta-2 agonists, but USMLE doesn't (i.e., some sources say tiotropium is recommended over ipratropium 1st-line now, but I've had students come out of the exam saying ipratropium showed up, but that it didn't matter since it was the only drug listed that worked anyway).
Oxybutynin	- Used for urge incontinence.
Scopolamine	- Scopolamine patches are used for motion sickness (i.e., put it on the arm before getting on a boat or plane). This drug is an anti-cholinergic straight-up. - As mentioned above, oral diphenhydramine can also be used for motion sickness, but scopolamine is the classic treatment.
Hyoscyamine	- Used sometimes for irritable bowel syndrome.
Non-depolarizing nicotinic receptor antagonists	
Tubocurarine	- "Non-depolarizing" means the agent does not cause any action potentials. - NBME Q mentions a non-depolarizing agent used for neuromuscular blockade during surgery, and the answer is just "tubocurarine" straight-up.
Vecuronium	- Used for paralysis during surgery. - Cisatracurium and rocuronium are other important agents.
Depolarizing nicotinic receptor agonist (acts as antagonist)	
Succinylcholine	- Functions as antagonist due to desensitization of receptors. - Binds to nicotinic receptors and agonizes them, causing initial action potentials and muscle twitching (fasciculations), prior to desensitization and muscle paralysis. - Akin to leuprolide causing desensitization of GnRH receptors with continued binding, where despite being a GnRH receptor agonist, it acts functionally as an antagonist. - Can cause malignant hyperthermia (high fever + muscular rigidity; treat with dantrolene, which inhibits the ryanodine Ca^{2+} channel).
Smoking cessation agents	
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Varenicline	- Used for smoking cessation. - Partial agonist at nicotinic receptors, but since this effect is less than endogenous ACh, it functions like an antagonist.

Epilepsy/hypnotic agents	
- USMLE doesn't care about MOAs for these. They care about side-effects mostly, as per my observation.	
Valproic acid	- Blocks sodium channels and ↑ GABA. - Used as an alternative to lithium in Tx of bipolar disorder. - Causes neural tube defects in pregnancy due to interference with folate metabolism.
Carbamazepine	- Blocks sodium channels. - Generic anti-epileptic; also used as prophylaxis for trigeminal neuralgia. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause aplastic anemia and SIADH.
Phenytoin	- Blocks sodium channels. - Generic anti-epileptic. - Causes neural tube defects in pregnancy due to interference with folate metabolism. - Can cause fetal hydrantoin syndrome (finger nail hypoplasia + abnormal facies). - Used for status epilepticus if benzo fails.
Ethosuximide	- Blocks thalamic calcium channels. Only anti-epileptic USMLE might ask MOA. - Used for absence seizures.
Benzodiazepines	- Diazepam, lorazepam, midazolam, chlordiazepoxide, etc. - Bind to GABA-A receptor, which ↑ frequency of Cl^- channel opening, thereby allowing Cl^- to flow into the neuron, hyperpolarizing it and inhibiting it (i.e., ↓ firing). - Hypnotic (induces sleep); anxiolytic (reduces anxiety); anti-convulsant. - Given 1st-line for status epilepticus. - Clonazepam used for insomnia/anxiety (shows up on NBME). - Used for stimulant intoxication (i.e., cocaine, amphetamine, PCP). - Used for specific phobia (e.g., fear of flying). - Used for social phobia 2nd-line if patient has asthma, since we don't give propranolol in this setting (asked on NBME). - Used for alcohol withdrawal (delirium tremens / alcoholic hallucinosis). - USMLE wants you to know that alcohol also binds to and activates GABA-A receptor, but at a different location from benzos. There's an NBME Q where you have to select the illustration of an alcohol molecule and benzo binding to the same receptor, but at different sites. - Can cause respiratory depression. - Treat overdose with flumazenil (antagonist of benzo receptor).
Barbiturates	- Phenobarbital, thiopental, etc. - Bind to GABA-A receptor; ↑ duration of chloride channel opening, making them more efficacious (and dangerous) than benzos. - Used for general anesthesia. - Used for unremitting status epilepticus if benzos and phenytoin fail.
Non-benzo hypnotics	- Zolpidem, zaleplon. - Activate GABA-A receptor, but bind to a different subunit from benzos. - Can cause addiction similar to benzos.
Magnesium	- Used for Tx and prevention of eclamptic seizures in pregnancy. - Given to pregnant women giving birth <32 weeks' gestation for neuroprotection for the fetus. - Used for torsades arrhythmia and digoxin toxicity.
Lamotrigine	- Blocks sodium channels. - General anti-epileptic. - Can cause Stevens-Johnson syndrome.

Drugs for neuropathic pain	
TCAs	- Prevent reuptake of both serotonin and norepinephrine. - Amitriptyline, nortriptyline, imipramine. - Neuro use-case is neuropathic pain, i.e., diabetic peripheral neuropathy. - Causes CCC → Coma, Convulsions, Cardiotoxicity (QT prolongation on ECG).

	- Have nasty anti-cholinergic side-effects (cognitive dysfunction and urinary retention, especially in elderly).
Gabapentin	- Blocks calcium channels, leading to neuroinhibition. - Used for neuropathic pain, e.g. , diabetic peripheral neuropathy.

Headache drugs	
Migraine	- Abortive therapy: NSAIDs first, then sumatriptan (serotonin receptor agonist). - Prevention: propranolol (beta-blockade).
Cluster	- Abortive therapy: oxygen. - Prevention: verapamil (non-dihydropyridine calcium channel blocker).
Tension	- Rest + acetaminophen.
Trigeminal neuralgia	- Abortive: goes away on its own because episodes last maximum 30-60 seconds. - Prevention: carbamazepine (sodium channel blocker).

Parkinson drugs	
Carbidopa-levodopa	- Combo frequently used for Tx of Parkinson disease. - Levodopa crosses the BBB to be converted to dopamine centrally. However, levodopa is subject to fast metabolism when administered alone. The addition of carbidopa functions as a competitive inhibitor of breakdown enzymes, resulting in increased levodopa availability for passage across the BBB. Do not confuse this mechanism with direct COMT inhibitors (tolcapone, entacapone), which prevent breakdown of L-dopa. - Carbidopa-levodopa can cause psychosis if administered in too-high a dose. This is assessed on 2CK Psych forms, where if patient gets psychotic episodes following recent addition of C-L to regimen, or following an increase in dose, the answer is simply "decrease dose of carbidopa-levodopa." "Discontinue carbidopa-levodopa is the wrong answer."
D2 receptor agonists	- Ropinirole, pramipexole, cabergoline, pergolide, and bromocriptine. - Used for Parkinson disease management. - Bromocriptine classic agent used for prolactinoma (dopamine agonism ↓ prolactin). - Ropinirole + pramipexole classically show up as restless leg syndrome treatment following making sure the patient is iron replete. New literature suggests gabapentin is first-line over D2 agonists now for RLS, but USMLE won't play trivia. They will give patient with RLS + say, "In addition to ropinirole, what else could be added?" And gabapentin will be the only answer listed that works.
Amantadine	- Used for Parkinson disease. - Increases presynaptic release of dopamine.
Selegiline	- Used for Parkinson disease. - Inhibits monoamine oxidase B, which is an enzyme that preferentially breaks down dopamine. - Can cause serotonin syndrome, either alone, or in combo with drugs like St John Wort or SSRIs.

Alzheimer drugs	
Donepezil	- As mentioned earlier, this is a cholinesterase inhibitor used in Alzheimer. - Rivastigmine and galantamine you can be aware of and have same MOA, but are LY.
Memantine	- NMDA glutamate receptor antagonist. - Glutamate receptor activation normally causes Ca^{2+} to flow into the neuron, leading to depolarization and neuroexcitation. In other words, USMLE wants you to know glutamate receptor is a ligand-gated calcium channel (where glutamate is the ligand that activates the channel). Antagonism causes neuroinhibition.

Quick anesthetics garbage	
MAC	- Mean alveolar concentration. - Used to determine the potency of inhaled anesthetics. - Dose of inhaled anesthetic required to prevent movement in 50% of patients to external noxious stimuli. - Less drug needed to achieve effect = more potent = lower MAC. - More drug needed to achieve effect = less potent = higher MAC. - Drugs that are more lipophilic (i.e., have high blood:lipid coefficient) are more potent and have lower MAC. This is because they dissolve well into the membranes of neurons and are able to achieve faster anesthesia. - Halothane = high potency = low MAC + high blood:lipid coefficient. - Nitrous oxide = low potency = high MAC + low blood:lipid coefficient.
Blood:lipid coefficient	- Reflects how lipophilic / how well the drug dissolves in adipose tissue. - As discussed above, high blood:lipid coefficient = lower MAC = higher potency. - Halothane is an example of a drug with high blood:lipid coefficient and high potency. - Nitrous oxide is an example of a drug with low blood:lipid coefficient and low potency. - USMLE wants you to know that coming out / waking up from anesthesia is due to “redistribution of drug into adipose tissue.
Blood:gas coefficient	- Reflects solubility in blood. - Low blood:gas coefficient = fast onset of action (nitrous oxide). - High blood:gas coefficient = slow onset of action (halothane). - Blood:gas and blood:lipid coefficients are not necessarily inversely related – i.e., drugs that have higher potency (high B:L coefficient) don’t necessarily have slow onset of action (high B:G coefficient).

Inhaled anesthetics	
- All used for general anesthesia (i.e., to put the patient to sleep). - All potentiate GABA + inhibit NMDA glutamate receptors.	
Halothane	- Can cause malignant hyperthermia (same as succinylcholine). - Rarely used today due to severe liver toxicity, but you need to know it as classic agent. - High potency due to high blood:lipid coefficient.
Nitrous oxide	- Fast-onset of action (low blood:gas coefficient).
Sevoflurane	- Fast-onset of action (low blood:gas coefficient).

Intravenous anesthetics	
Ketamine	- NMDA glutamate receptor antagonist. - Often used in trauma situations for procedural sedation.
Propofol	- Classic agent used to maintain general anesthesia. - White in color (looks like milk). - Has calories. (Won’t get asked on USMLE, but weird factoid anesthetists sometimes ask). - The drug that killed Michael Jackson.
Benzos	- Midazolam is classic agent used for adjunct to anesthesia as an anxiolytic.
Barbiturates	- Thiopental used classically for induction of anesthesia but has been replaced by propofol.

Local anesthetics	
- Block sodium channels.	
Lidocaine	- Very common local anesthetic. - Also can be used as a type Ib anti-arrhythmic in theory.
Procaine	- Short duration of action (i.e., effect doesn't last long).
Bupivacaine	- Long duration of action (i.e., effect lasts long).

Neurotoxic drugs	
Vincristine	- Prevents microtubule assembly; used as general chemotherapy. - USMLE can ask what drug inhibits and answer is just "tubulin." - Peripheral / autonomic neuropathy. - Answer shows up as "toxic neuropathy" on a 2CK Neuro CMS form.
Cisplatin	- Crosslinking agent used as general chemotherapy. - Causes neuro- and ototoxicity.
Aminoglycosides	- 30S ribosomal subunit inhibitors used for gram-negative rods. Highest yield use-case is in combination with vancomycin as empiric Tx for endocarditis. - Gentamicin, tobramycin, amikacin (latter sounds unusual but shows up on an NBME). - Cause ototoxicity (sensorineural hearing loss and/or vertigo). - Cause nephrotoxicity.
Loop diuretics	- Block the $1Na^+/1K^+/2Cl^-$ symporter on the apical membrane of the thick ascending limb. - Furosemide, ethacrynic acid, etc., can cause sensorineural hearing loss and/or vertigo.
Aspirin	- Irreversible cyclooxygenase 1 and 2 inhibitor. - Most common symptom is tinnitus (ear-ringing).
Quinidine	- 1a sodium channel blocker used occasionally as anti-arrhythmic. - Causes cinchonism, which means headache + tinnitus.
Digoxin	- Na^+/K^+-ATPase inhibitor used for advanced heart failure. - Optic neuritis presenting as yellow wavy, "Vincent van Gogh vision."
Ethambutol	- Arabinosyl transferase inhibitor used for TB treatment. - Causes optic neuritis (visual changes of any kind, e.g., blurry vision, central scotoma, red-green color blindness).
Sildenafil	- Phosphodiesterase-5 inhibitor used for erectile enhancement (Viagra). - Can cause blue haze. Apparently this is due to its ability to also inhibit PDE-6 found in retinal receptors (holy shit). Tadalafil (Cialis) doesn't inhibit PDE-6 the same way, so this effect is more just with Viagra. - Step 3 wants you to know that pilots cannot take this within 3 hours of getting on an airplane due to possible blue haze. Unusual/obscure factoid, as if one would require a boner before getting in the cockpit, no pun intended.

If you've read the above, you've finished this PDF for all intents and purposes for USMLE. You can use my HY Neuroanatomy PDF if you want more in depth neuro.

I'm adding the original bullet points from my old HY Neuro PDF below because some whiny med students said they preferred this style instead. Hard to appease everyone.

- 33F + 1-wk Hx of R-sided weakness + L-sided inability to feel temperature/pain + diminished vibratory sensation over R foot + brisk R-sided reflexes + R-sided Babinski sign; Dx? → answer on NBME is Brown-Sequard syndrome secondary to viral infection.
- 59F + metastatic cancer + in pain + crying + "wants to die"; Q asks most likely reason for wanting to die; answer = "inadequate pain control"; "major depression" is wrong answer; must address pain management in cancer patients.
- 42F + 3-month Hx of insomnia + discomfort while lying in bed; next best step in management? → check serum iron and ferritin levels; student says wtf? → restless leg syndrome is most often caused by iron deficiency.
- 42F + 3-month Hx of insomnia + discomfort while lying in bed + serum iron and ferritin are normal; next best step in management? → D2 agonist – i.e., pramipexole or ropinirole, etc.
- Patient with restless leg syndrome is at increased risk for what disease later in life? → answer on USMLE = Parkinson disease (if D2 agonist can Tx, then lack of dopamine transmission may be etiology in some patients).
- 58M + loses consciousness while shaving + tilt-table test shows **no abnormalities**; Dx? → NBME wants "carotid sinus hypersensitivity" as answer. If tilt-table test (+), answer = vasovagal syncope.
- 45F + fundoscopy shows hard exudates + cotton wool spots + scattered hemorrhages; Dx? → diabetic retinopathy.
- Medication that can cause tardive dyskinesia that is not an antipsychotic? → answer = metoclopramide (D2 antagonist); can also prolong QT interval and cause hyperprolactinemia.
- Anti-depressant med causing seizures? → bupropion.
- Frontal lobe injury in car accident; NBME asks which deficit is most likely to ensue; answer = conceptual planning.

- 56M + alcoholism + acutely intoxicated + B1 is administered; the latter decreases what most significantly? → Neuro shelf wants "anterograde amnesia" as the answer; mnemonic for Wernicke = A COW → Ataxia, Confusion, Ophthalmoplegia, Wernicke.
- 56M + 3-day Hx of cutting from 12 beers a day down to 4; develops tremulousness; Tx? → chlordiazepoxide (delirium tremens); classic vignette is guy has surgery + two days later has tachycardia, tremulousness, and hallucinations (alcoholic hallucinosis).
- 50F + high ESR and creatine kinase (CK) + proximal muscle weakness +/- muscle pain; Dx? → polymyositis.
- 50F + high ESR + no mention of high CK or weakness in the vignette + muscle pain + muscle stiffness; Dx? → polymyalgia rheumatica (PMR).
- Main difference between PMR and polymyositis? → PMR has no proximal muscle weakness + a normal creatine kinase; polymyositis has high CK and weakness; pain + stiffness of muscles can be seen in both conditions, but classically PMR. For USMLE vignettes + neuro shelf, focus on whether there's weakness or elevated CK.
- 48F + high ESR and creatine kinase (CK) + proximal muscle weakness +/- muscle pain; next best step in Dx? → Neuro shelf wants either "anti-Jo1 / -Mi2 antibodies" or "electromyography and nerve conduction studies" as the answer to Dx polymyositis. After these, do muscle biopsy can be performed for definitive Dx. In contrast, no specific Dx test is used for PMR.
- 59F + temporal headache + muscle pain and stiffness + high ESR; Dx? → temporal arteritis + polymyalgia rheumatica → next best step? = IV methylprednisolone first, followed by temporal artery biopsy.
- 68M + Hx of prostate cancer + now has neurologic findings; next best step in Dx? → MRI of spine to look for mets → **however if corticosteroids are listed, choose these before MRI** → for spinal cord compression on NBME (from cancer or trauma), choose steroids first if listed.
- 65F + breast cancer + neurologic findings; next best step? → answer on Neuro NBME is "intravenous high-dose dexamethasone."
- 72M + prostate cancer + neurologic findings; Dx? → answer = "epidural spinal cord compression" due to spinal mets → do steroids first (if listed), then MRI.

- Patient has brain cancer; next best step? → steroids.
- How to Dx brain cancer? → answer = contrast head CT (done for cancer and abscess).
- How to Dx brain bleed? → answer = non-contrast CT (always done for intracranial bleeds).
- Epidural hematoma; next best step after CT confirms? → answer = **intubation and hyperventilation** on Neuro NBME. After this is done, craniotomy is the answer.
- Subdural hematoma; next best step after CT? → Answer = craniotomy on NBME, not observation.
- What do you see on non-contrast CT with epidural vs subdural? → epidural hematoma = lens-shaped bleed; subdural = crescent-shaped.
- Blood pressure of 220/120 + confusion; Dx + Tx? → Hypertensive encephalopathy; give IV sodium nitroprusside → this question is asked on Neuro form 1.
- BP of 220/120 + sodium nitroprusside administered; now patient has confusion; Dx? → cyanide toxicity caused by nitroprusside.
- Drug that can be given to prevent vasospasm after a subarachnoid hemorrhage (SAH)? → nimodipine (dihydropyridine CCB).
- Severe headache + stiff neck; Dx? → SAH (can cause meningism, similar to meningitis).
- Brain bleed in patient with Alzheimer; Dx? → amyloid angiopathy (intracerebral hemorrhage).
- 87F + Alzheimer + low-grade fever + delirium; next best step? → answer = do urinalysis to look for UTI as cause of delirium.
- Viral infection + tinnitus + vertigo +/- neurosensory hearing loss → labyrinthitis.
- Viral infection + vertigo → vestibular neuritis.
- Tx for acute flare of MS → IV steroids (oral is wrong and can make flares worse).
- Tx between flares of MS (the patient must by asymptomatic) → IFN-beta (interferon beta).
- Tx for spasticity in MS → baclofen (GABA-B receptor agonist).
- Incontinence in MS → urge (hyperactive detrusor; detrusor instability).
- Mechanism for MS? → T cell-mediated attack against oligodendrocytes.
- How to Dx MS? → MRI is gold standard; choose MRI over CSF IgG oligoclonal bands.
- 27F + intermittent headaches + blurry vision; Dx? → optic neuritis (multiple sclerosis) → student says “why the headaches?” Yeah, I know. Weird. But it’s on the NBME. You need to know optic neuritis is

HY in MS and means inflammation of cranial nerve II → presents as blurry vision, or change in color vision, or Marcus Gunn pupil (relative afferent pupillary defect).

- Most specific eye finding in MS → medial longitudinal fasciculus (MLF) syndrome → aka internuclear ophthalmoplegia (INO) → when you abduct to one side, you activate CN VI on that side, which requires the contralateral CN III to activate in order to adduct → the side that cannot adduct is the side that's fucked up; the normal side will have nystagmus.
- How to differentiate CN III lesion from INO? → INO patients can converge normally.
- 20F + stiffness of hands + frontal balding + impaired relaxation of hypothenar muscles; Dx? → myotonic dystrophy (CTG TNR disorder).
- Cancer patient on cisplatin or vincristine + develops neuropathy; Dx? → answer = toxic neuropathy (chemotherapy-induced neuropathy); these drugs are neurotoxic.
- Confusion and/or seizure due to electrolyte disturbance; Dx? → hypo-/hypernatremia and hypercalcemia all cause CNS dysfunction.
- Confusion in the setting of high BUN and creatinine; Dx? → uremic encephalopathy.
- Floppy baby syndrome; Dx and location of CNS/PNS affected? → Werdnig-Hoffmann syndrome (spinal muscular atrophy); anterior horns affected.
- Antipsychotic medication started + muscle rigidity + no fever; Dx + Tx? → acute dystonia, not neuroleptic malignant syndrome (because no fever); Tx with benztropine (muscarinic receptor antagonist) or diphenhydramine (1st gen H1 blocker, which has strong anti-muscarinic side-effects).
- Antipsychotic medication started + abnormal eye movements + stiff neck; Dx? → acute dystonia (oculogyric crisis + torticollis).
- Antipsychotic medication started + muscle rigidity + fever; Dx + Tx? → neuroleptic malignant syndrome; give dantrolene (inhibits ryanodine channel).
- Antipsychotic med + restlessness; Dx + Tx? → akathisia; Tx with propranolol.
- Antipsychotic med + bradykinesia; Dx + Tx? → drug-induced Parkinsonism; Tx with amantadine or propranolol.
- Antipsychotic med + abnormal tongue movements; Dx + Tx? → tardive dyskinesia; stop antipsychotic + switch to atypical.
- Tx for diabetic neuropathic pain? → answer = TCA (i.e., amitriptyline). Second-line is gabapentin

- Tx for herpetic / post-herpetic neuralgia (i.e., from shingles)? → gabapentin
- 82M diabetic + neuropathic pain + already taking carbamazepine + gabapentin to no avail; next best step? → switch the meds to nortriptyline (a TCA) → student then asks, "Wait, I thought you said TCAs are first-line. Why does this Q have the guy on those two meds then?" → two points: 1) we don't like giving TCAs to elderly because of their anticholinergic and anti-alpha-1 side-effects, so this vignette happen to try other agents first, but if you're asked first-line, always choose TCA; and 2) if we do give a TCA to an elderly patient, we choose nortriptyline because it carries fewer adverse effects.
- How to differentiate cluster headache from trigeminal neuralgia? → cluster will be a male 20s-40s with 11/10 lancinating pain behind the eye waking him up at night (he may pace around the room until it goes away); details such as lacrimation and rhinorrhea are too easy and will likely not show up on the shelf. In contrast, trigeminal neuralgia will be 11/10 lancinating pain behind the eye (or along the cheek / jaw if V2 or V3 branches affected; it's when V1 is affected that this diagnoses are more readily confused) → TN is brought on by a minor stimulus such as brushing one's hair or teeth, or a gust of wind.
- Tx and prophylaxis for cluster headache? → Tx = 100% oxygen; prophylaxis = **verapamil.**
- Tx and prophylaxis for trigeminal neuralgia? → Tx = goes away on its own because it lasts only seconds; prophylaxis = **carbamazepine.**
- Tx and prophylaxis for migraine? → Tx = NSAID, followed by triptan (triptans are NOT prophylaxis; they are for abortive therapy only after NSAIDs); prophylaxis = **propranolol.**
- 32M + diffuse headache relieved by acetaminophen + sleep; Dx? → answer = tension-type headache; Tx = rest + taper caffeine (if taking it).
- Other HY uses for propranolol?
    - Migraine prophylaxis (FM form gives patient with HTN + migraine; answer = propranolol)
    - Akathisia (with antipsychotic use)
    - Thyroid storm (decreases peripheral conversion of T4 to T3)
    - Essential tremor (bilateral resting tremor in young adult; autosomal dominant; patient will self-medicate with EtOH, which decreases tremor).
    - Hypertension + idiopathic tremor (i.e., tremor need not be essential if patient has HTN → answer on FM form is "beta-adrenergic blockade" for the HTN Tx).

  - Esophageal varices prophylaxis (patients at risk of bleeds)
  - Hypertrophic obstructive cardiomyopathy (increases preload → decreases murmur)
  - Social phobia

- 47M + bitemporal hemianopsia; most likely tumor? → prolactinoma (most common pituitary tumor in adults).
- 8M + bitemporal hemianopsia; most likely tumor? → craniopharyngioma (most common pituitary tumor in children; some sources argue not true pituitary tumor because it's derived from Rathke pouch, which is the roof of the primitive pharynx).
- Most common primary brain cancer in children? → pilocytic astrocytoma → solid + cystic on MRI; has Rosenthal fibers → not aggressive.
- 3F + morning vomiting + truncal ataxia; Dx? → medulloblastoma of cerebellar vermis.
- Most common primary brain cancer in adults? → glioblastoma multiforme (butterfly glioma) → areas of necrosis + hemorrhage (large, irregular mass on head CT); often presents with seizure.
- 44F + SLE + irregular ring-enhancing lesion seen on head CT; Dx? → primary CNS lymphoma, not Toxo.
- Neuro shelf asks Tx for Toxo; answer? → sulfadiazine + pyrimethamine; prophylaxis is trimethoprim + sulfamethoxazole. For PJP, TMP/SMX is Tx and prophylaxis. Yes, this is asked on Neuro shelf.
- Neonate + intracranial calcifications + chorioretinitis + hydrocephalus; Dx? → Congenital Toxo.
- Neonate + intracranial calcifications + deafness + hepatomegaly + rash; Dx? → CMV.
- Neonate with bilateral deafness due to maternal infection + no other info given; Dx? → answer = congenital CMV.
- CNS tumors seen in neurofibromatosis? → oligodendroglioma + ependymoma + **optic nerve glioma** in NF1; in NF2, meningioma. These may overlap, but be aware that these tumors can merely occur in either NF, period.
- 50F + jaw pain + headaches + normal ESR; Dx? → temporal mandibular joint syndrome; if ESR is high, answer is jaw claudication caused by temporal arteritis.
- 40F being treated for TB + has neurologic findings in extremities +/- seizure; Dx? → B6 (pyridoxine) deficiency caused by isoniazid.
- 65F + metastatic breast cancer + suprapubic mass + decreased sphincter tone + pain over lumbar vertebrae on fist percussion; Dx? → answer = metastases to cauda equina.

- 44M + saddle anesthesia + late urinary incontinence; Dx? → cauda equina syndrome → bundle of spinal nerves and rootlets arising from the conus medullaris).
- 44M + perianal anesthesia + early urinary + fecal incontinence; Dx? → conus medullaris syndrome → bulbous end of the spinal cord = conus medullaris.
- 48M + burns his thumb; pain sensed via which dermatome? → answer = C6; lateral digits primarily C6; middle finger C7; medial digits primarily C8.
- 13F + 2-day Hx of lower limb paresthesias + CSF protein 95 mg/dL + CSF leukocyte count 4/uL; Dx? → answer = Guillain-Barre syndrome (GBS). USMLE will often not mention preceding infection, especially at 2CK level.
- What does CSF show in GBS? → albuminocytologic dissociation (high protein but normal cells).
- 24M + weakness proximally and distally in lower limbs + weakness distally in upper limbs; rest of the vignette is vague; Dx? → GBS → this descriptor means paralysis/paresis has ascending in lower limbs but not yet in upper limbs.
- Dx of Guillain-Barre syndrome? → Neuro shelf wants "electromyography and nerve conduction studies" over CSF findings.
- Mechanism of GBS? → T cell attack against Schwann cells of PNS.
- Tx of GBS? → IVIG + plasmapheresis; steroids are the wrong answer.
- 32F + paresthesias in thenar region of hand +/- hand weakness + sensation intact over dorsum of hand; next best step in Dx? → Neuro NBME exam wants "Electrophysiological testing." Call it weird, but it's what they want. Examination findings such as Tinel sign, Phalen maneuver, Flick test are insufficient for diagnosis.
- First Tx for carpal tunnel syndrome in patient who can't stop offending activity (e.g., office worker) → **wrist splint first;** then **triamcinolone (steroid) injection into the carpal tunnel**; do not select anything surgical as it's always wrong on the USMLE; NSAIDs are a wrong answer and not proven to help
- Tx for cubital tunnel syndrome → elbow splint.
- What is cubital tunnel syndrome → ulnar nerve entrapment at elbow → presents similarly to carpal tunnel syndrome but just in an ulnar distribution and involves the forearm.
- What is Guyon canal syndrome → ulnar nerve entrapment at the wrist → **hook of hamate fracture** or chronic handle bar impaction in avid cyclists.

- Midshaft fracture of humerus; which nerve is injured? → radial → wrist-drop + pronated forearm.
- Supracondylar fracture of humerus; which nerve is injured? → median.
- Medial epicondylar fracture; nerve injury? → ulnar.
- Surgical neck of humerus fracture; nerve injury? → axillary.
- Mastectomy + post-operative winged scapula; nerve injury? → long thoracic nerve.
- Which muscle is innervated by long thoracic nerve? → serratus anterior.
- 6M + ECG shows miscellaneous arrhythmia + seizure-like episode; Dx? → Adam-Stokes attack → not true seizure disorder as per EEG; arrythmia leads to hypoxia of brainstem → seizure-like fits ensue.
- 75M + episodes of loss of consciousness (LoC) for 2 years + tonic-clonic-like episodes + becomes pale and sweaty + Hx of MI; Dx? → answer = "syncope" on the NBME (convulsive syncope).
- Mechanism of narcolepsy? → answer = deficiency of orexin (hypocretin).
- Diagnosis of narcolepsy? → answer = polysomnography.
- What is cataplexy? → loss of muscle tone usually in response to emotional stimulus (e.g., laughter) → seen in narcolepsy.
- Tx of nacrolepsy = stimulants (i.e., modafinil).
- What is simple vs complex seizure? → simple = no LoC; complex = LoC; patient staring off into space not aware of surroundings = LoC.
- What is partial vs generalized seizure? → partial = affecting one part of the brain; generalized = involves the wholes of both cerebral hemispheres.
- What is myoclonic seizure? → jerks in muscle or group of muscles; no LoC.
- Empiric Abx therapy for meningitis → ceftriaxone + vancomycin (+/- steroids).
- Lumbar puncture or Abx first in suspected meningitis? → new guidelines say LP first.
- When do you do CT head before LP in suspected meningitis?
    - Confusion that interferes with neurologic exam / decreased GCS score.
    - Seizure.
    - Focal neurologic signs (motor or sensory).
    - Papilledema or if the optic fundi cannot be visualized.
    - Above reasons indicate potential mass lesion, where if you do an LP you can cause tonsillar herniation and death; if CT negative, proceed cautiously to LP.

- Bacterial meningitis: low glucose, high protein, high neutrophils (polymorphonuclear cells; PMNs).
- Aseptic (viral) meningitis: normal glucose, normal (or slightly elevated) protein, high lymphocytes.
- Fungal meningitis: low glucose, high protein, high lymphocytes (similar to bacterial, but high lymphocytes instead of neutrophils).
- Herpes encephalitis: **lots of RBCs in CSF due to temporal lobe hemorrhage** → CT is often negative, but sometimes Q will mention wave slowing or temporal complexes on EEG. Tx = IV acyclovir.
- Difference between meningitis and encephalitis → meningitis is nuchal rigidity (neck stiffness) + photophobia + ophthalmoplegia; encephalitis presents with **confusion.**
- Dx of Cryptococcal meningitis? → answer = **latex agglutination** if it's listed over India ink; mucicarmine staining (red stain) can also be done.
- Tx for Cryptococcal meningitis → amphotericin B + flucytosine, followed by fluconazole taper.
- Nodular density in upper lobe in immunocompromised pt → aspergilloma → next best step = open lung biopsy (sounds radical, but it's the answer on one of the NBME forms) → Tx with -azole → invasive aspergillosis can be treated with caspofungin or voriconazole.
- 44F + diplopia + dysphagia + eyelid ptosis; all worsen throughout the day; Dx? → myasthenia gravis (MG).
- 44F + proximal muscle weakness + able to perform upward gaze without a problem for 60 seconds; Dx? → Lambert-Eaton (LE) syndrome (if cannot perform, answer = MG).
- Vignette where Dx is either MG or LE but it's not listed; answer? → "neuromuscular junction."
- 44F + difficulty getting up from chair but is able to after several attempts; Dx? → LE.
- Mechanism of MG vs LE? → MG = antibodies against postsynaptic acetylcholine receptors; LE = antibodies against presynaptic voltage-gated calcium channels.
- MG can sometimes be paraneoplastic syndromes of which cancer? → MG from thymoma (do chest imaging to check for thymoma after Dx of MG; if thymoma present + removed, this cures the MG). Up to 80% of patients with thymoma + MG have anti-MUSK Abs (muscle-specific kinase).
- LE can sometimes be a paraneoplastic syndrome of which cancer? → small cell lung cancer.
- How to Dx MG vs LE? → If both are listed, choose antibodies over Tensilon (edrophonium) test.
- What is edrophonium → short-acting acetylcholinesterase inhibitor.
- How do MG vs LE perform with edrophonium? → MG improves drastically; LE less change.

- Tx of MG? → give an acetylcholinesterase inhibitor (i.e., pyridostigmine).
- 38F + papilledema on fundoscopy + flushing of face / venous distension of head + neck; Dx? → NBME wants "obstruction of venous return to the heart" → SVC syndrome; classically has (+) Pemberton sign; why this is on Neuro shelf I do not know, but it is. Tangentially, thymic lesions can cause SVC-like syndrome (i.e., T cell ALL in peds)
- 6M + dancing eyes + HTN + lesion visualized in posterior mediastinum on CXR; Dx? → neuroblastoma; students says wtf? → can occur anywhere in the median sympathetic chain, although classically intra-abdominal. Dancing eyes = opsoclonus-myoclonus syndrome.
- Neuroblastoma; how to Dx? → answer on NBME is "urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA); mIBG scan may also be used; n-myc gene.
- 25M + impaired plantar flexion + inversion; Dx? → tibial nerve injury.
- 25M + impaired dorsiflexion + eversion; Dx? → common peroneal nerve injury.
- 22F returns from Central America + has tonic-clonic seizure + CT of head shows "swiss cheese" appearance of brain, or cystic structures in ventricles, or white specs somewhat diffusely (I've seen all three images on NBME forms); Dx? → neurocysticercosis (Taenia solium); Tx = albendazole (or praziquantel).
- AIDS patient with CD4 count of 47/uL + cognitive degeneration + miscellaneous peripheral neurologic findings; Dx? → progressive multifocal leukoencephalopathy (PML) caused by JC polyoma virus; USMLE wants "reactivation of latent infection"; "acute infection in immunocompromised patient" is the wrong answer. T2-weighted MRI shows multiple hyperintensities.
- 55F + few-month Hx of spinning sensation + no vomiting + when head hangs off examination table she develops coarse, rotary nystagmus beating to one side; Dx? → answer on NBME is benign paroxysmal positional vertigo (BBPV); need not be associated with vomiting; Dx with Dix-Hallpike maneuver; Tx with Epley maneuver.
- Cause of BPPV? → posterior semicircular canal **otolith.**
- What is cholesteatoma? → non-cancerous squamous proliferation of middle ear.
- Most common cause of carotid plaques? → HTN → the strong systolic impulse from the heart pounds the carotids --> endothelial damage --> atherosclerosis.

- 55M + BP 150/90 + TIA; next best step in Mx? → carotid duplex USS → the first thing you want to think about is, "does this guy have a carotid plaque that has resulted in a clot embolizing to his brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA; next best step in Mx? → ECG → you want to think, "Does he have atrial fibrillation with a LA mural thrombus that's now embolized to the brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA + ECG shows sinus rhythm with no abnormalities; next best step in Mx? → Holter monitor → when you first see this scenario you're probably like, "Wait, the ECG is normal, so it's not AF?" → No, *it is* likely AF, but AF is often paroxysmal, so in order to detect it in this scenario, **the next best step is a Holter monitor (24-hour wearable ECG).** This means that later in the day when he sits down to have dinner and then pops into AF, the Holter monitor will pick it up.
- What % of people over age 80 have AF? → 8% of people over age 80 have AF, which is why age is a huge risk factor. In other words, if the vignette says the guy is 58, AF is probably less likely just based on shear probability, regardless of hypertensive status." And, once again, knowing that AF is often paroxysmal is really important.
- Age 50s-60s + high BP + TIA/stroke/retinal artery occlusion; next best step in Dx? → answer = carotid duplex ultrasound to look for carotid plaques.
- Age >75 + good BP + TIA/stroke/retinal artery occlusion; answer = ECG to look for AF → if normal, do Holter monitor to pick up paroxysmal AF.
- 55M + good BP + carotid bruit heard on auscultation; next best step in Mx? → answer = carotid duplex ultrasound to look for carotid plaques → in this case, if they are obvious and explicit about the suspected etiology of the stroke, TIA, or retinal artery occlusion, then you can just do the carotid duplex ultrasound.
- How to Mx carotid plaques? → first we have to ask whether the patient is symptomatic or asymptomatic. A bruit does not count as symptoms (that's a sign). Symptomatic means stroke, TIA, or retinal artery occlusion. According to recent guidelines: carotid occlusion >70% if symptomatic, or >80% if asymptomatic → answer = do carotid endarterectomy. Below these thresholds → answer = medical management = statin, PLUS clopidogrel OR dipyridamole + aspirin. The USMLE will actually

not be hyper-pedantic about the occlusion %s (that's Qbank). They'll make it obvious for you which answer they want. They'll say either 90% → answer certainly = carotid endarterectomy, or they'll say 50% → answer = medical management only. There's one NBME Q where they say a guy has a bruit but is asymptomatic, and has 10 and 30% occlusion in the left vs right carotids, respectively, and he's already on aspirin + statin, and the answer is "maintain current regimen" → if he were symptomatic, even with low occlusion, he'd certainly need statin, PLUS clopidogrel OR dipyridamole + aspirin.

- 6F + rheumatic fever + has Sydenham chorea; Q asks mechanism → answer = autoimmune disorder to the basal ganglia.
- 22F + says cannot see out of left eye + examination shows no relative afferent pupillary defect; answer = conversion disorder – i.e., if he truly had impaired vision, Marcus-Gunn pupil would be present.
- 38F + hoarseness of voice + various stress factors; Dx? → conversion disorder; merely be aware that hoarseness of voice is assessed for this.
- Back pain worse when standing or walking for long periods of time → lumbar spinal stenosis.
- Radiculopathy down an arm → cervical disc herniation.
- Bilateral paresthesias in the arms in rheumatoid arthritis patient → atlantoaxial subluxation.
- Bilateral paresthesias in the arms in rheumatoid arthritis patient → MRI of spine to Dx atlantoaxial subluxation.
- Prior to surgery in rheumatoid arthritis patient → cervical CT or flexion/extension x-rays of cervical spine to check for atlantoaxial subluxation.
- 40M + combination of LMN and UMN findings + **no** sensory abnormalities; Dx? → amyotrophic lateral sclerosis; make sure you memorize that there are NO sensory findings, as this is the detail most students forget. Sometimes the answer won't be "ALS," but will instead be "motor neuron" (i.e., they'll ask for structural diagnosis, rather than disease name).
- 1-year-old boy + hypotonia + honey consumption; Dx? → answer = ingestion of spores (honey); botulism (*Clostridium botulinum*).
- 13M + camping trip + Bell palsy + constipation + hypotonia; Dx? → ingestion of preformed toxin from compromised canned goods. Tx = anti-toxin.

- 32M + waxing and waning tinnitus and hearing loss + difficulty hearing conversations at dinner and in groups + has family Hx of similar Sx; Dx? → Meniere disease → low-frequency hearing loss.
- Mechanism for Meniere? → defective endolymphatic drainage; may attempt antihistamines and diuretics.
- Viral infection + tinnitus +/- vertigo + no hearing loss; self-resolves over weeks to months; Dx? → vestibular neuritis.
- Viral infection + tinnitus + hearing loss +/- vertigo self-resolves over weeks to months; Dx? → labyrinthitis. One Q on the neuro forms has presentation similar to vestibular neuritis but answer is labyrinthitis; vestibular neuritis isn't listed as alternative answer.
- 3M + pinna displaced upward and outward + fever; Dx? → mastoiditis (malignant otitis externa).
- 3M + mastoiditis + next best step in Mx? → CT or MRI of the temporal bone (x-ray is wrong answer) → sounds outrageous to do a CT in a kid, but "CT of temporal bone" is answer on one of the NBMEs; in UWorld, MRI is answer + x-ray is wrong/insufficient → mastoiditis often associated with fluid collection that must be visualized + drained to prevent brain abscess.
- 25M IV drug user + Tx for endocarditis + now the room is spinning; Dx? → CN VIII toxicity caused by aminoglycoside (can be hearing loss, tinnitus, or vertigo); empiric endocarditis Tx is usually gentamicin + either vancomycin or ampicillin/sulbactam.
- Endocarditis patient + has stroke-like episode + fever; next best step? → IV antibiotics → septic embolus to the brain; can lead to brain abscess.
- MCA stroke on dominant hemisphere (usually left-sided); HY findings? → contralateral limb + face sensory and/or motor deficits; also classically associated with Broca and Wernicke aphasias.
- ACA stroke; HY findings? → contralateral lower leg sensory and/or motor dysfunction.
- PCA stroke; HY findings? → contralateral hemianopsia with macular sparing.
- Broca aphasia; HY findings? → non-fluent aphasia; telegraphic speech; comprehends normally but cannot communicate easily, leading to frustration; **repetition impaired.**
- Where is Broca area? → posterior, lateral, inferior **frontal lobe.**
- Wernicke aphasia; HY findings? → fluent aphasia; word salad; non-sensical speech + does not comprehend; **repetition impaired.**
- Where is Wernicke area? → superior temporal gyrus (**temporal lobe**).

- What is conduction aphasia? → patient has only impaired repetition (arcuate fasciculus connecting Broca and Wernicke areas).
- What is global aphasia? → patient has presentation of Broca and Wernicke aphasias at the same time.
- What is transcortical motor aphasia? → patient sounds like he/she has Broca aphasia but **repetition is intact.**
- What is transcortical sensory aphasia? → patient sounds like he/she has Wernicke aphasia but **repetition is intact.**
- What is mixed transcortical aphasia? → patient sounds like he/she has Broca and Wernicke at the same time but **repetition is intact.**
- 42M + acalculia + left-right agnosia + finger agnosia + agraphia; Dx? → Gerstmann syndrome (angular gyrus syndrome; junction of parietal and temporal lobes).
- Stroke causing Horner syndrome + dysphagia + loss of pain/temp from contralateral body and ipsilateral face; Dx? → lateral medullary syndrome (PICA infarct) → "Pikachu" = PICA-chew = dysphagia with PICA infarct.
- Stroke causing ipsilateral Bell palsy; Dx? → lateral pontine syndrome (AICA infarct) → FACIAL spelled backwards has AICA in it → face is affected.
- Stroke causing **ipsilateral tongue deviation** + contralateral paralysis and loss of propioception; Dx? → medial medullary syndrome (paramedian branch of anterior spinal artery).
- Lacunar infarct HY points? → caused by HTN leading to lipohyalinosis of lenticulostriate vessels → if affecting posterior limb of internal capsule, deficits are pure motor (contralateral); if thalamic, deficits are pure sensory (contralateral).
- Stroke causing locked-in syndrome; Dx? → Basilar artery stroke.
- Correct hyponatremia too quickly? → locked-in syndrome due to central pontine myelinolysis (osmotic demyelination).
- Correct hypernatremia too quickly? → cerebral edema.
- Pain in contralateral limb months after stroke resolves; Dx? → thalamic pain syndrome.
- Coma or persistent vegetative state following deceleration injury (i.e., MVA or fall); Dx? → diffuse axonal injury.

- 7F + facial grimaces past 5 months + no other motor findings or abnormal sounds + mental status normal; next best step in Mx? → answer = "schedule a follow-up examination in 3 months" → Dx = provisional tic disorder → 1/5 children experience some form of tic disorder; most common ages 7-12; usually lasts less than a year; "watch and wait" approach recommended. Provisional tic disorder used to be called transient tic disorder; the name was changed because a small % go on to develop chronic tics.
- 10-month old boy + jerking movements of the limbs + EEG shows chaotic high-amplitude spikes with no recognizable pattern; Dx? → answer = West syndrome (infantile spasms) → triad of spasms + hypsarrhythmia on EEG (no recognizable pattern with high amplitude spikes) + developmental regression; international definition of the diagnosis requires two out of three; starts age 3-12 months; seen in 1-5% of Down syndrome kids; Tx = **ACTH,** prednisolone, or vigabatrin; ACTH is thought to act by two mechanisms: 1) stimulates steroid (cortisol) release; 2) direct (cortisol-independent) effect on melanocortin receptors.
- 13M + tonic-clonic seizure + 4-month Hx of hypnagogic/hypnopompic jerking of left arm + uncle has epilepsy; Dx? → answer = juvenile myoclonic epilepsy; genetic with unclear inheritance pattern; characterized by myoclonic jerks (usually hypnagogic and/or hypnopompic) that progress to tonic-clonic seizures after several months; age of onset is usually 10-16, but can also start in adulthood; Tx is valproic acid.
- 4F + few-month Hx of near-daily seizures + seizures typically occur while she's sleeping + has started putting objects in her mouth and making less eye contact + seizures not responding to anti-epileptic meds; Dx? → answer = Lennox-Gastaut syndrome → severe childhood-onset epilepsy characterized by near-daily seizures and cognitive decline (hyperoralism is a sign of cognitive regression [babies put things in their mouths]); poor prognosis, with 5% mortality rate in childhood; 80-90% persistence of seizures into adulthood.



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